

Graph Neural Networks for Earthquake Early Warning: A Comprehensive Review of Methods, Applications, and Future Directions

Ahmad R. Purnama^{a,b,*}, Wei Zhang^a, Hiroshi Tanaka^b, Maria C. Rodriguez^c

^aDepartment of Computer Science, Faculty of Engineering, Universitas Diponegoro, Semarang, Indonesia

^bDepartment of Earth Sciences, Faculty of Natural Sciences, Universitas Diponegoro, Semarang, Indonesia

^cInstitute of Geophysics, National Seismological Center, Santiago, Chile

Abstract

Earthquake Early Warning (EEW) systems provide seconds to tens of seconds of advance notice before destructive ground shaking arrives at a target site. Traditional approaches rely on empirical scaling relationships and threshold-based detection algorithms that exhibit well-documented limitations in accuracy, speed, and robustness, particularly for large-magnitude events and sparse network geometries. Graph Neural Networks (GNNs) offer a paradigm shift by natively representing seismic monitoring networks as graphs, where stations constitute nodes and spatial relationships define edges. This graph-structured representation enables explicit modeling of inter-station correlations, wavefield propagation patterns, and network topology effects that are fundamental to seismological analysis but difficult to capture with conventional architectures.

This review systematically examines GNN methodologies for EEW, covering architectures from spectral methods (GCN) to spatial approaches (GAT, GraphSAGE) and temporal-spatial formulations. We analyze applications spanning seismic phase picking, event detection, source characterization, magnitude estimation, ground motion prediction, and source localization. Detailed comparisons against traditional methods (STA/LTA, template matching) and deep learning baselines (PhaseNet, EQTransformer) demonstrate that GNN approaches achieve measurable improvements, particularly for sparse networks and complex multi-event sequences. We examine benchmark datasets (STEAD, INSTANCE, LEN-DB), identify critical challenges including real-time processing constraints, transfer learning across tectonic regions, and model interpretability, and outline future directions encompassing physics-informed GNNs, foundation models for seismology, federated learning, and edge computing deployment. We give particular attention to the rapid 2024–2026 progress toward deployment-oriented and generalizable systems—including graph-based continuous ground-motion prediction, unified multi-task picking–association–location networks, contrastive and universal models, and emerging seismic foundation models. This review provides researchers and practitioners with a comprehensive roadmap for advancing GNN-based EEW systems toward operational deployment.

Keywords: Graph Neural Networks, Earthquake Early Warning Systems, Seismology, Deep Learning, Seismic Network, Ground Motion Prediction, Phase Picking, Spatio-temporal Modeling

*Corresponding author

Email address: ahmad.purnama@ft.undip.ac.id (Ahmad R. Purnama)

1. Introduction

Earthquakes remain among the most devastating natural hazards, causing significant loss of life, infrastructure damage, and economic disruption worldwide. The 2011 Tohoku earthquake (M_w 9.0) in Japan resulted in over 15,000 fatalities and triggered a catastrophic tsunami (Simons et al., 2011), while the 2023 Turkey-Syria earthquake sequence (M_w 7.8) caused more than 50,000 deaths across a wide region. These events underscore the persistent vulnerability of human populations to seismic hazards and highlight the critical importance of effective early warning systems that can provide precious seconds of advance notice before destructive shaking arrives.

Earthquake Early Warning (EEW) systems exploit the differential propagation velocities of seismic waves—specifically, the faster but less destructive P-waves (compressional, ~ 6 km/s) versus the slower but more damaging S-waves (shear, ~ 3.5 km/s) and surface waves—to issue alerts before the arrival of strong ground motion at a target site (Allen and Kanamori, 2003; Allen, 2012; Allen and Melgar, 2019). The concept, first proposed in the late 1860s and operationalized in Japan following the development of the UrEDAS system (Nakamura, 1988), has matured into deployed systems across multiple countries including Japan (JMA), Mexico (SASMEX), the United States (ShakeAlert), Taiwan (CWA), Italy, and others (Gasparini et al., 2007; Given et al., 2018; Hsiao et al., 2009).

Despite decades of development, existing EEW systems face fundamental limitations. Traditional approaches based on empirical scaling relationships and threshold algorithms (e.g., τ_c - P_d methods) struggle with: (1) the inherent trade-off between speed and accuracy in source characterization; (2) magnitude saturation for large events when using only initial P-wave seconds; (3) high false alert rates in noisy environments; and (4) poor performance in regions with sparse station coverage (Satriano et al., 2011; Bose et al., 2018). These limitations have motivated the seismological community to explore data-driven approaches that can potentially overcome the constraints of physics-based empirical models.

1.1. The Rise of Deep Learning in Seismology

The application of deep learning to seismological problems has accelerated dramatically since 2018, building on foundational advances in representation learning (LeCun et al., 2015; Krizhevsky et al., 2012; He et al., 2016) and sequence modeling (Hochreiter and Schmidhuber, 1997; Graves et al., 2013; Schuster and Paliwal, 1997). Modern deep learning training relies on key techniques including adaptive optimizers (Kingma and Ba, 2015), regularization methods (Srivastava et al., 2014; Ioffe and Szegedy, 2015), and generative models (Goodfellow et al., 2014; Kingma and Welling, 2014). These advances enabled transformative progress in seismic phase picking (Ross et al., 2018; Zhu and Beroza, 2019; Mousavi et al., 2020), event detection (Perol et al., 2018; Yoon et al., 2015), earthquake location (Kriegerowski et al., 2019; Lomax et al., 2019), and ground motion prediction (Derras et al., 2014). Architectures such as PhaseNet (Zhu and Beroza, 2019) and EQTransformer (Mousavi et al., 2020) have demonstrated performance comparable to or exceeding manual analyst picks across diverse tectonic settings, as quantified by standardized benchmarking frameworks such as SeisBench (Woollam et al., 2022) and comparative picker evaluations (Munchmeyer et al., 2022). Comprehensive reviews by Bergen et al. (2019), Kong et al. (2019), Mousavi and Beroza (2022), and Mousavi and Beroza (2023) document the breadth of machine

learning applications in solid Earth geoscience, establishing deep learning as an indispensable tool in modern seismology.

However, most existing deep learning approaches treat seismic stations independently—processing waveforms from individual stations or station triplets through CNNs or RNNs without explicitly modeling the spatial relationships between stations in a network. This station-independent processing paradigm ignores a fundamental aspect of seismological observation: seismic monitoring networks are inherently graph-structured systems where the spatial configuration of stations, their inter-station distances, and the wavefield propagation patterns across the network carry essential information about earthquake sources.

1.2. Why Graph Neural Networks for Seismology?

Graph Neural Networks (GNNs) provide a natural computational framework for processing data defined on irregular graph structures (Scarselli et al., 2009; Bronstein et al., 2017; Wu et al., 2021; Zhang et al., 2022b). Recent theoretical perspectives on message passing architectures (Velickovic, 2022) further clarify their representational capabilities. In the context of seismic monitoring, a GNN represents stations as nodes equipped with recorded waveform features, while edges encode spatial relationships (distance, azimuth, geological similarity) between station pairs. The message-passing mechanism intrinsic to GNNs enables each station to aggregate information from neighboring stations, effectively learning how seismic wavefield characteristics evolve across the network topology.

This graph-based representation offers several compelling advantages for EEW applications:

- (i) **Network-aware processing:** GNNs naturally incorporate the full network geometry, enabling joint processing of multi-station observations with explicit spatial reasoning about wavefield propagation.
- (ii) **Permutation invariance:** GNN architectures are invariant to station ordering, making them robust to network configuration changes (station additions, removals, or failures).
- (iii) **Scalability:** GNNs can handle networks of varying sizes without architectural modifications, accommodating both dense urban arrays and sparse regional networks.
- (iv) **Dynamic adaptability:** Temporal graph formulations can capture the time-evolving nature of seismic wavefield propagation across station networks.
- (v) **Interpretability:** Attention-based GNN variants can identify which station connections are most informative for specific predictions, providing physically meaningful explanations.

Recent pioneering work has demonstrated the potential of GNNs for seismological applications, including earthquake source characterization (van den Ende and Ampuero, 2020), spatiotemporal event analysis (Zhang et al., 2021), earthquake location and magnitude estimation (McBrearty and Beroza, 2023a), and phase association (Munchmeyer et al., 2021; McBrearty and Beroza, 2023b). These studies consistently report improvements over station-independent methods, validating the hypothesis that network-level information processing enhances seismic analysis. Since 2024, this momentum has accelerated toward operationally oriented systems: graph-based continuous ground-motion prediction for

networks of arbitrary size (Clements et al., 2024), unified multi-task networks that jointly pick, associate, and locate (Si et al., 2024), and self-supervised contrastive GNNs that estimate shaking intensity from only a few seconds of data at a handful of stations (Murshed et al., 2024). In parallel, generalized and universal training strategies are being developed to make a single model transferable across monitoring setups (Zhang and Zhang, 2024), signaling a transition from proof-of-concept studies to deployment-ready GNN-based EEW.

1.3. Scope and Contributions of This Review

This review provides the first comprehensive survey dedicated to the intersection of Graph Neural Networks and Earthquake Early Warning systems. While previous reviews have covered machine learning in seismology broadly (Mousavi and Beroza, 2023; Bergen et al., 2019; Kong et al., 2019) or GNN methods generally (Wu et al., 2021; Zhou et al., 2020), none has specifically examined how graph-based deep learning architectures can address the unique challenges of EEW. Our contributions are:

1. A systematic taxonomy of GNN architectures and their applicability to different EEW sub-tasks, from phase detection through alert generation.
2. A critical comparative analysis of GNN-based approaches versus traditional and CNN-based methods across standardized benchmarks.
3. Identification of key research gaps and technical challenges specific to deploying GNNs in operational EEW contexts.
4. A forward-looking discussion of emerging paradigms—foundation models, physics-informed GNNs, federated learning—that may shape the next generation of EEW systems.

1.4. Paper Organization

The remainder of this review is organized as follows. Section 2 presents the fundamentals of earthquake early warning systems, including operational principles, deployed systems worldwide, and technical challenges. Section 3 discusses the representation of seismic networks as graph-structured data. Section 4 provides a comprehensive overview of graph neural network architectures relevant to seismology. Section 5 reviews GNN applications across the EEW processing pipeline. Section 6 examines temporal-spatial graph network formulations. Section 7 presents comparative analyses with traditional approaches. Section 8 surveys benchmark datasets and evaluation protocols. Section 9 discusses challenges and limitations. Section 10 outlines future research directions. Section 11 addresses ethical and operational considerations. Section 12 provides conclusions and recommendations.

2. Fundamentals of Earthquake Early Warning Systems

2.1. Principles of Earthquake Early Warning

The physical basis of earthquake early warning exploits the fundamental difference in propagation velocities between electromagnetic signals (traveling at the speed of light through communication networks)

and seismic waves (traveling at kilometers per second through the Earth’s crust). When an earthquake initiates, P-waves radiate outward at approximately 6–8 km/s in crustal rocks, followed by S-waves at approximately 3.5–4.5 km/s. The destructive ground shaking that causes most structural damage is primarily associated with S-waves and surface waves. An EEW system detects the initial P-wave arrival at one or more stations, rapidly estimates the earthquake’s source parameters (location, magnitude, and predicted ground motion), and transmits an alert to target sites before the arrival of damaging waves (Allen and Kanamori, 2003; Satriano et al., 2011).

The warning time Δt_{warn} available at a target site is determined by:

$$\Delta t_{\text{warn}} = \frac{d_{\text{target}}}{v_S} - \frac{d_{\text{station}}}{v_P} - t_{\text{proc}} - t_{\text{comm}} \quad (1)$$

where d_{target} is the distance from source to target, d_{station} is the distance from source to the detecting station, v_P and v_S are the P- and S-wave velocities respectively, t_{proc} is the data processing time, and t_{comm} is the communication latency. Minimizing t_{proc} while maintaining accuracy is the central algorithmic challenge in EEW.

Two fundamental approaches to EEW have been developed:

- **On-site (single-station) methods:** These utilize the initial P-wave recorded at a single station near the target to predict the subsequent S-wave intensity at that same location. Methods include the τ_c - P_d approach (Wu and Kanamori, 2005) and the predominant period method. While providing minimal warning time, they are applicable even in regions with sparse instrumentation.
- **Regional (network-based) methods:** These combine observations from multiple stations to determine source parameters (epicenter, depth, magnitude) and predict ground motion at distant target sites using ground motion prediction equations (GMPEs). Regional methods generally provide larger warning times for distant targets but require denser station networks and more complex algorithms (Allen, 2012; Gasparini et al., 2007).

The EEW processing pipeline comprises several sequential tasks: (1) real-time data acquisition and quality control; (2) P-wave detection and onset time picking; (3) event association (determining which detections belong to the same earthquake); (4) source location estimation; (5) magnitude estimation from initial waveform parameters; (6) ground motion prediction at target sites; and (7) alert dissemination with decision logic for issuing public warnings.

2.2. Operational EEW Systems Worldwide

Several countries have deployed operational EEW systems, each reflecting different network densities, tectonic settings, and performance requirements. Table 1 summarizes the major operational systems.

Japan’s system, operated by the Japan Meteorological Agency (JMA), represents the most mature operational EEW deployment, covering the entire Japanese archipelago with over 1,100 stations (Hoshiba and Ozaki, 2015). The system demonstrated both its capabilities and limitations during the 2011 Tohoku earthquake, where the initial magnitude estimate significantly underestimated the true M_w 9.0

Table 1: Summary of operational Earthquake Early Warning systems worldwide. Station counts reflect approximate current deployments.

System	Country	Year	Stations	Method	Alert Time
UrEDAS/JMA	Japan	2007	~1,100	Network	5–30 s
SASMEX	Mexico	1991	~97	Regional	60–75 s
ShakeAlert	USA	2021	~1,675	Network	5–20 s
CWA	Taiwan	2014	~700	Regional/On-site	10–20 s
PRESTo	Italy	2009	~200	Regional	5–15 s
EEWS	South Korea	2015	~314	Network	5–25 s
CISN/ElarmS	California	2006	~400	Network	3–10 s
READI	Romania	2013	~150	Regional	20–30 s

magnitude—a phenomenon known as magnitude saturation that remains a critical challenge for EEW systems processing large events.

Mexico’s SASMEX system (Espinosa-Aranda et al., 1995) exploits the favorable geometry of the country’s subduction zone, where large earthquakes nucleate along the Pacific coast far from Mexico City, providing up to 60–75 seconds of warning time. The ShakeAlert system in the western United States (Given et al., 2018) uses the ElarmS, EPIC, and FinDer algorithms to process data from over 1,675 stations across California, Oregon, and Washington, delivering public alerts since October 2019. Italy’s PRESTo system (Zollo et al., 2014) demonstrates the applicability of regional EEW in the complex tectonic environment of the Apennines.

Taiwan’s system (Hsiao et al., 2009; Wu and Kanamori, 2005) benefits from the island’s high seismicity and dense station coverage, achieving typical alert times of 10–20 seconds. The system combines both regional and on-site approaches, with recent upgrades incorporating machine learning algorithms for improved phase picking and magnitude estimation.

2.3. Technical Challenges in Existing EEW Systems

Despite decades of development and demonstrated success, current EEW systems face several persistent technical challenges that motivate the exploration of novel data-driven approaches:

Speed-accuracy trade-off: The fundamental tension between processing speed and estimation accuracy represents the defining challenge of EEW. Using only 1–3 seconds of P-wave data, traditional methods achieve magnitude uncertainties of ± 0.5 –1.0 magnitude units, which translates to order-of-magnitude uncertainties in predicted ground motion intensity (Satriano et al., 2011). Extending the processing window improves accuracy but reduces the available warning time.

Magnitude saturation: Empirical scaling relationships between initial P-wave parameters (pre-dominant period τ_c , peak displacement P_d) and final magnitude tend to saturate for events above M_w 7.0 (Wu and Kanamori, 2005). This saturation arises because the finite fault rupture duration exceeds the short processing window, causing the system to underestimate the true magnitude of the largest—and most dangerous—events.

False alert management: Maintaining low false alert rates while preserving high detection sensitivity is critical for public trust. Non-earthquake signals (teleseisms, cultural noise, sensor glitches, telemetry errors) can trigger false alerts. Japan’s system, for example, issued a false M_w 6.6 alert in January 2018 due to the simultaneous arrival of signals from two separate smaller earthquakes at nearby stations.

Sparse network coverage: Many seismically active regions have insufficient station density for reliable network-based EEW. This is particularly problematic in developing countries where seismic risk is highest but monitoring infrastructure is limited. Algorithms must perform robustly with observations from only a few stations.

Real-time constraints: EEW algorithms must complete all processing within 1–3 seconds of the first P-wave detection. This severe latency constraint limits the computational complexity of applicable methods and has historically precluded the use of sophisticated machine learning models that require significant inference time.

2.4. Evolution from Rule-based to Data-driven Approaches

The evolution of EEW algorithms mirrors the broader transformation occurring across geophysical monitoring. First-generation EEW systems relied on deterministic rules: fixed STA/LTA thresholds for detection (Allen, 1978), empirical τ_c - P_d scaling for magnitude estimation, and predetermined Ground Motion Prediction Equations (GMPEs) for ground motion forecasting (Abrahamson et al., 2014; Boore et al., 2014).

Second-generation approaches introduced statistical frameworks, Bayesian estimation, and ensemble methods that improved uncertainty quantification while maintaining real-time performance (Satriano et al., 2011). The Bayesian approach enables the integration of prior seismicity information and the progressive updating of source parameter estimates as additional data become available.

Third-generation systems, emerging since approximately 2018, increasingly incorporate deep learning components. Initial applications focused on replacing individual modules—using CNNs for phase picking (Ross et al., 2018; Zhu and Beroza, 2019) or magnitude estimation—while maintaining the overall pipeline architecture. The recognition that seismic networks constitute natural graph structures has motivated the development of fourth-generation approaches based on Graph Neural Networks that can potentially replace the entire processing pipeline with end-to-end learned models (van den Ende and Ampuero, 2020; McBrearty and Beroza, 2023a).

3. Seismic Networks as Graph-Structured Data

3.1. Network Topology and Station Connectivity

A seismic monitoring network can be formally represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the set of N seismic stations (nodes), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ defines the connectivity structure (edges), and $\mathbf{X} \in \mathbb{R}^{N \times F}$ represents the node feature matrix with F features per station. This representation naturally captures the irregular spatial distribution of seismic stations—unlike regular grid structures assumed by CNNs—and enables explicit modeling of inter-station relationships.

The graph structure encodes essential physical information about how seismic waves propagate between stations. Stations that are geographically proximate sample similar portions of the wavefield and share correlated signals, while distant stations may observe the same event from fundamentally different perspectives (azimuth, distance, site conditions). The graph representation allows GNNs to learn these spatially varying relationships directly from data.

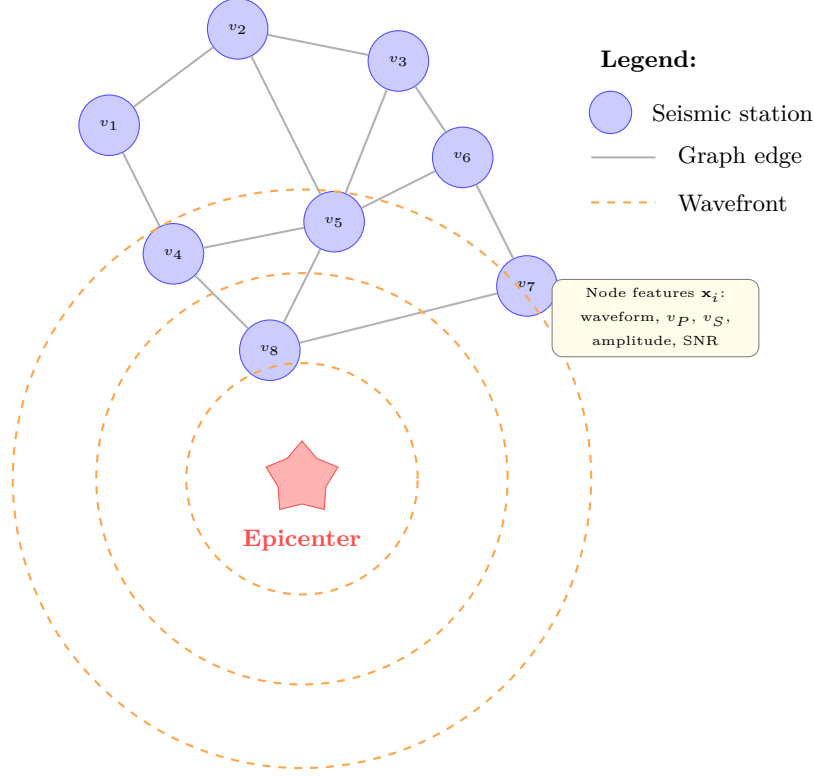


Figure 1: Illustration of a seismic monitoring network represented as a graph. Stations (nodes) are connected by edges encoding spatial relationships. Seismic wavefronts propagate outward from the earthquake epicenter, arriving at stations at different times depending on distance. The graph structure enables GNNs to model inter-station correlations and wavefield coherence.

3.2. Graph Construction Strategies

The choice of graph construction strategy—how to define the edge set $\mathcal{E}$ and associated edge weights—significantly influences GNN performance. Several approaches have been explored in the seismological context:

Distance-based graphs: The most common approach connects stations within a threshold distance $d_{\max}$:

$$e_{ij} = \begin{cases} 1 & \text{if } \text{dist}(v_i, v_j) \leq d_{\max} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Edge weights are typically computed using a Gaussian kernel: $w_{ij} = \exp(-\text{dist}(v_i, v_j)^2/\sigma^2)$, where σ controls the distance decay. This approach captures the physical intuition that nearby stations share similar ground motion characteristics, with the correlation decaying with distance (van den Ende and Ampuero, 2020; McBrearty and Beroza, 2023a).

k -Nearest Neighbor (k -NN) graphs: Each station is connected to its k nearest neighbors, ensuring a minimum connectivity regardless of station density:

$$\mathcal{N}(v_i) = \arg \text{top-}k_{v_j \in \mathcal{V} \setminus \{v_i\}} [-\text{dist}(v_i, v_j)] \quad (3)$$

This approach handles the heterogeneous station densities common in real networks, where urban areas may have stations separated by less than 10 km while rural regions have inter-station distances exceeding 100 km (Zhang et al., 2021).

Voronoi tessellation-based graphs: The Delaunay triangulation of station positions defines a natural planar graph that captures the geometric neighborhood structure. This approach produces edges that respect the spatial tessellation of the monitoring area and has been used in studies requiring physically interpretable connectivity (Yano et al., 2021).

Learned graph structures: Rather than defining graph topology a priori, some approaches learn the adjacency matrix jointly with the GNN parameters. This allows the model to discover non-obvious station correlations—potentially reflecting shared geological structures, similar site conditions, or common noise sources—that may not correspond to geographic proximity (Bai et al., 2020; Wu et al., 2020).

3.3. Node and Edge Feature Engineering for Seismic Data

The effectiveness of GNN-based seismic processing depends critically on the choice of node and edge features that encode relevant physical information.

Node features ($\mathbf{x}_i \in \mathbb{R}^F$) typically include:

- Raw or preprocessed waveform segments (3-component: vertical, north, east)
- Spectral features: predominant period τ_c , frequency content, spectral ratios
- Amplitude measures: peak displacement P_d , peak velocity, peak acceleration, cumulative absolute velocity (CAV)
- Temporal features: P-wave arrival time relative to origin time, time since first detection
- Station metadata: elevation, site classification (V_{S30}), instrument type, noise level
- Signal quality indicators: signal-to-noise ratio (SNR), data completeness

Edge features ($\mathbf{e}_{ij} \in \mathbb{R}^D$) encode pairwise station relationships:

- Inter-station distance and azimuth
- Differential arrival times: $\Delta t_{ij} = t_i^P - t_j^P$
- Cross-correlation coefficients between waveforms at connected stations
- Geological path effects (velocity structure along the inter-station path)
- Relative site amplification factors

3.4. Dynamic vs Static Graph Representations

A critical design decision for GNN-based EEW is whether the graph structure should remain fixed (static) or evolve over time (dynamic).

Static graphs maintain a fixed adjacency matrix throughout processing, based on pre-computed station geometry. This approach is computationally efficient and suitable for tasks where the network configuration does not change during an event. Most existing work adopts static graphs (van den Ende and Ampuero, 2020; McBrearty and Beroza, 2023a).

Dynamic graphs allow the graph structure to evolve as an earthquake unfolds. As P-waves propagate outward from the source, stations are progressively “activated” (incorporated into the graph) as they detect signals. The graph at time t includes only stations that have recorded arrivals:

$$\mathcal{V}(t) = \{v_i \in \mathcal{V} : t_i^{\text{detect}} \leq t\} \quad (4)$$

This temporal evolution naturally captures the expanding spatial footprint of seismic observations and is particularly relevant for EEW, where decisions must be made with progressively increasing but initially sparse network observations (Münchmeyer et al., 2022). Dynamic graph approaches align with the sequential decision-making nature of EEW, where magnitude and location estimates are refined as additional stations report detections.

Hybrid approaches combine static backbone connectivity with dynamic node activation, maintaining edge definitions between all station pairs but only performing message passing through activated nodes. This formulation preserves computational efficiency while capturing the temporal evolution of network observations during an earthquake.

4. Graph Neural Network Fundamentals

4.1. Message Passing Framework

The unifying computational paradigm underlying most modern GNN architectures is the message passing neural network (MPNN) framework (Gilmer et al., 2017), which generalizes earlier gated graph sequence models (Li et al., 2016). In this framework, each node iteratively updates its representation by aggregating information from its neighbors through a sequence of message passing layers. A theoretical analysis of the expressive power of such architectures, connecting them to the Weisfeiler-Leman graph isomorphism test, is provided by Morris et al. (2019) and Xu et al. (2019). For a node v_i with hidden representation $\mathbf{h}_i^{(l)}$ at layer l , the update is:

$$\mathbf{m}_i^{(l+1)} = \bigoplus_{j \in \mathcal{N}(i)} \phi(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{e}_{ij}) \quad (5)$$

$$\mathbf{h}_i^{(l+1)} = \psi(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l+1)}) \quad (6)$$

where ϕ is the message function computing interactions between connected nodes, $\bigoplus$ is a permutation-invariant aggregation operator (sum, mean, or max), and ψ is the update function that combines the aggregated messages with the node’s current state. The initial node representations are $\mathbf{h}_i^{(0)} = \mathbf{x}_i$.

After L layers of message passing, each node’s representation $\mathbf{h}_i^{(L)}$ encodes information from its L -hop neighborhood—the set of all nodes reachable within L graph traversal steps. For EEW applications, this receptive field determines how far across the seismic network information can propagate, with implications for the model’s ability to capture network-wide wavefield patterns.

For graph-level predictions (e.g., earthquake magnitude from the entire network), a readout function aggregates all node representations:

$$\mathbf{h}_{\mathcal{G}} = \text{READOUT} \left(\{ \mathbf{h}_i^{(L)} : v_i \in \mathcal{V} \} \right) \quad (7)$$

4.2. Spectral Methods

Spectral graph neural networks define convolution operations through the graph Fourier transform, leveraging the eigendecomposition of the graph Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ is the adjacency matrix.

ChebNet (Defferrard et al., 2016) approximates spectral graph convolutions using Chebyshev polynomials T_k of the normalized Laplacian $\tilde{\mathbf{L}} = 2\mathbf{L}/\lambda_{\max} - \mathbf{I}$:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}}) \mathbf{h}_i^{(l)} \right) \quad (8)$$

where K determines the polynomial order (localization radius) and θ_k are learnable parameters. This formulation enables localized filtering without computing the full eigendecomposition, achieving $O(K|\mathcal{E}|)$ complexity.

Graph Convolutional Network (GCN) (Kipf and Welling, 2017) simplifies ChebNet by using first-order approximation ($K = 1$) with renormalization:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right) \quad (9)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ includes self-loops, $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$, and $\mathbf{W}^{(l)}$ is a learnable weight matrix. GCN has become the most widely adopted baseline architecture for graph learning tasks, including initial explorations in seismology (Bilal et al., 2022; Bloemheuveld et al., 2022).

For seismic applications, spectral methods offer the advantage of well-understood frequency-domain properties but are limited to fixed graph topologies (the Laplacian eigendecomposition is graph-specific), which restricts their applicability to dynamic network scenarios.

4.3. Spatial Methods

Spatial GNN methods define convolution directly in the vertex domain, operating through neighborhood aggregation without requiring spectral decomposition.

GraphSAGE (Hamilton et al., 2017) introduces sampling-based neighborhood aggregation for inductive learning:

$$\mathbf{h}_{\mathcal{N}(i)}^{(l)} = \text{AGG} \left(\{ \mathbf{h}_j^{(l-1)} : j \in \mathcal{N}(i) \} \right) \quad (10)$$

$$\mathbf{h}_i^{(l)} = \sigma \left(\mathbf{W}^{(l)} \cdot \left[\mathbf{h}_i^{(l-1)} \parallel \mathbf{h}_{\mathcal{N}(i)}^{(l)} \right] \right) \quad (11)$$

where AGG is an aggregation function (mean, LSTM, or max-pooling) and $\parallel$ denotes concatenation. GraphSAGE’s inductive capability—learning generalizable aggregation functions rather than node-specific embeddings—is particularly valuable for seismology, where models trained on one network region must generalize to other regions with different station configurations.

Graph Attention Network (GAT) (Velickovic et al., 2018) introduces attention mechanisms to learn adaptive weighting of neighbor contributions:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))} \quad (12)$$

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j^{(l)} \right) \quad (13)$$

Multi-head attention extends this by concatenating H independent attention heads. For EEW, the attention mechanism provides physically interpretable weights: stations contributing more informative signals (higher SNR, closer to the epicenter, better azimuthal coverage) naturally receive higher attention weights, and these weights can be visualized to understand model decision-making (van den Ende and Ampuero, 2020).

Graph Isomorphism Network (GIN) (Xu et al., 2019) achieves maximal expressiveness among message-passing GNNs by using summation aggregation with a learnable parameter ϵ :

$$\mathbf{h}_i^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) \cdot \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right) \quad (14)$$

GIN’s theoretical expressiveness makes it suitable for tasks requiring fine discrimination between different graph structures—relevant when distinguishing seismic events with subtly different source characteristics observed through similar network geometries.

4.4. Temporal Graph Networks

Seismic data is inherently temporal, requiring architectures that jointly capture spatial (inter-station) and temporal (within-waveform) patterns.

T-GCN (Zhao et al., 2020) combines Graph Convolutional layers for spatial processing with Gated Recurrent Units (GRUs) for temporal modeling:

$$\mathbf{f}^{(t)} = \text{GCN}(\mathbf{X}^{(t)}, \mathbf{A}) \quad (15)$$

$$\mathbf{h}^{(t)} = \text{GRU}(\mathbf{f}^{(t)}, \mathbf{h}^{(t-1)}) \quad (16)$$

This architecture processes spatial correlations at each time step, then captures temporal dynamics through the recurrent mechanism.

A3T-GCN extends T-GCN with attention mechanisms that adaptively weight temporal observations, allowing the model to focus on the most informative time steps—such as the initial P-wave onset or the transition to S-wave arrival (Bai et al., 2020).

DCRNN (Diffusion Convolutional Recurrent Neural Network) (Li et al., 2018) models spatial dependencies through bidirectional random walks on the graph, integrated with sequence-to-sequence learning

for temporal prediction:

$$\mathbf{H}^{(l)} = \sum_{k=0}^K \left(\theta_{k,1} (\mathbf{D}_O^{-1} \mathbf{A})^k + \theta_{k,2} (\mathbf{D}_I^{-1} \mathbf{A}^T)^k \right) \mathbf{X} \quad (17)$$

where $\mathbf{D}_O$ and $\mathbf{D}_I$ are the out-degree and in-degree matrices. The diffusion process naturally captures wave propagation patterns on the station graph.

Spatio-Temporal Graph Convolutional Networks (STGCN) (Yu et al., 2018) interleave temporal convolution (1D causal convolutions along the time axis) with spatial graph convolution, avoiding the sequential processing bottleneck of RNN-based approaches while maintaining receptive field growth through dilated convolutions.

4.5. Graph Transformers

Recent work has extended the Transformer architecture (Vaswani et al., 2017) to graph-structured data, combining the expressiveness of global self-attention with structural graph information.

Graphormer (Ying et al., 2021) incorporates graph structure through spatial encodings (shortest-path distances), edge encodings (features along shortest paths), and centrality encodings (node degree information) into the standard Transformer attention mechanism. This allows attention to be computed between all node pairs while remaining sensitive to graph topology.

Graph Transformer architectures (Dwivedi and Bresson, 2021) adapt positional encodings for graphs using Laplacian eigenvectors or random walk probabilities as positional features. For seismic networks, these positional encodings can capture the geometric arrangement of stations relative to potential source locations.

Graph Transformers offer the advantage of global receptive fields (every node can attend to every other node) without the depth limitations of message-passing networks, which may be beneficial for large-scale seismic networks where wavefield information must propagate across many hops.

4.6. Pooling and Readout Operations

For graph-level predictions in EEW (e.g., estimating a single magnitude value from the entire network), node representations must be aggregated into a fixed-size graph representation.

Global pooling operations (sum, mean, max) provide simple permutation-invariant aggregation:

$$\mathbf{h}_G = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{h}_i^{(L)} \quad (18)$$

Hierarchical pooling methods (Ying et al., 2018) learn to progressively coarsen the graph, identifying clusters of stations that can be summarized together—potentially discovering meaningful spatial groupings (e.g., stations on similar geological units, stations at similar epicentral distances).

Attention-based pooling (Lee et al., 2019) computes importance scores for each node and performs weighted aggregation, effectively learning which stations are most informative for a given prediction task. In EEW, stations closer to the epicenter with clear P-wave onsets would naturally receive higher attention weights.

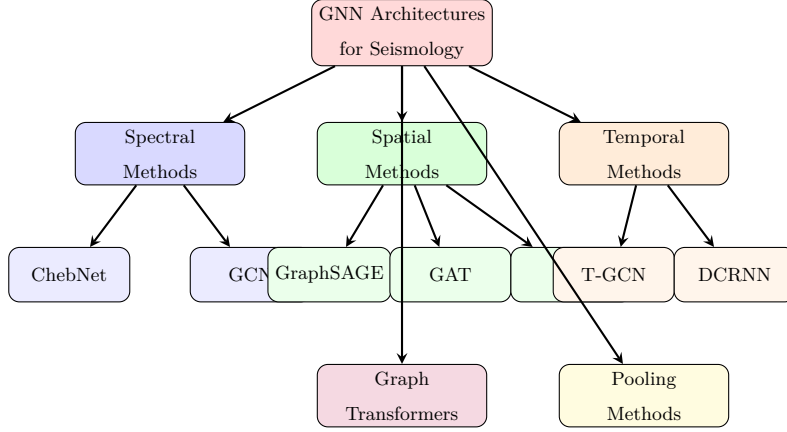


Figure 2: Taxonomy of Graph Neural Network architectures applicable to seismological data processing. Methods are categorized into spectral approaches (operating in the graph frequency domain), spatial approaches (aggregating in the vertex domain), temporal approaches (combining spatial and temporal modeling), graph transformers (global attention with structural encoding), and pooling methods (graph-level representation learning).

5. GNN Applications in Seismology

5.1. Seismic Phase Picking and Event Detection

Phase picking—the identification of P-wave and S-wave arrival times on seismograms—constitutes the foundational task in seismic data processing. Accurate phase picks are prerequisites for earthquake location, magnitude estimation, and tomographic imaging. Traditional methods including STA/LTA ratio algorithms (Allen, 1978; Withers et al., 1998) and manual analyst picks, while well-established, face limitations in throughput, consistency, and performance in low-SNR environments.

The application of GNNs to phase picking represents a paradigm shift from station-independent processing to network-aware detection. Bilal et al. (2022) demonstrated that GNN-based phase detection incorporating multi-station spatial correlations achieves superior performance compared to single-station CNN methods, particularly for small-magnitude events where individual station recordings may be ambiguous. Their approach constructs a graph over the seismic network and applies GAT layers to aggregate spatial context, enabling the model to leverage correlated arrivals at multiple stations for more confident phase identification.

Munchmeyer et al. (2021) applied GNNs to the closely related task of phase association—determining which detected phases belong to the same seismic event. This combinatorial problem, which scales quadratically with the number of detections in traditional grid-search methods, is naturally formulated as edge classification on a detection graph where nodes represent phase picks and edges connect potentially associated picks. The GNN learns to predict edge labels (same-event or different-event) by aggregating features from neighboring picks, achieving significant improvements in association accuracy for complex sequences with overlapping events. Earlier deep learning approaches to phase association include PhaseLink (Ross et al., 2019b), while Bayesian methods such as GAMMA (Zhu et al., 2022) provide complementary probabilistic frameworks. Scalable workflow systems such as QuakeFlow (Zhu et al., 2023a) integrate these association methods into end-to-end monitoring pipelines. A particularly influential graph-based associator is the Graph Earthquake Neural Interpretation Engine (GENIE) of

McBrearty and Beroza (2023b), which represents the station geometry and the candidate source region as two coupled graphs and learns, from data, to jointly localize sources and assign arrivals to them—robustly handling variable station counts and high pick rates. Pushing this integration further, Si et al. (2024) proposed PLAN, an all-in-one graph network that performs phase picking, association, and location simultaneously for multi-task, multi-station monitoring, exploiting the shared structure among these traditionally separate stages so that information learned for one task reinforces the others.

Bloemheuveld et al. (2022) proposed a graph-based framework for seismic phase classification that jointly models spatial and temporal relationships between waveform segments across the network. By operating on graphs where nodes carry short waveform windows and edges encode spatial proximity, the model achieves robust classification even in scenarios with multiple simultaneous events or high cultural noise levels. Moving beyond fixed connectivity, Piriyaatit et al. (2025) introduced a spatio-temporal graph convolutional network with a spectral structure-learning convolution that learns both static and dynamic inter-station relationships directly from multi-station waveforms, producing station-specific detection probabilities and illustrating the benefit of letting the graph topology itself be inferred rather than prescribed.

5.2. Earthquake Source Characterization

Earthquake source characterization encompasses the estimation of source parameters including hypocenter location, focal mechanism (fault orientation and slip direction), rupture geometry, and stress drop. van den Ende and Ampuero (2020) presented one of the earliest applications of deep GNNs to seismic source characterization, demonstrating that a GNN operating on the station network graph can directly estimate moment tensor components from multi-station waveform recordings. Their architecture processes three-component waveforms at each station through CNN feature extractors, then aggregates across stations using graph neural network layers, producing a single moment tensor estimate for the network. The approach showed particular advantages for events recorded by sparse networks, where the spatial reasoning enabled by the GNN compensates for limited azimuthal coverage.

Zhang et al. (2021) extended this work with spatiotemporal graph convolutional networks that jointly model the time-evolving wavefield across the station network. Their architecture interleaves temporal convolutions (capturing waveform evolution at each station) with spatial graph convolutions (aggregating across stations), achieving improved characterization of complex ruptures where the source time function evolves over multiple seconds. Further developments include deep deconvolution methods for source characterization using GNNs (van den Ende and Ampuero, 2022), graph-based seismic event classification frameworks (Zhang et al., 2022a), and graph neural networks applied to multivariate seismic time series (Bloemheuveld et al., 2022).

5.3. Magnitude Estimation

Rapid magnitude estimation from initial P-wave seconds represents one of the most critical EEW tasks, as magnitude directly determines predicted ground motion intensity and thus alert decisions. Traditional approaches using empirical scaling relationships between initial waveform parameters (τ_c , P_d) and final magnitude suffer from well-documented saturation effects for large events.

McBrearty and Beroza (2023a) demonstrated that GNNs can estimate earthquake magnitude with significantly reduced uncertainty compared to single-station methods by leveraging multi-station P-wave observations through graph-based processing. Their approach achieves magnitude estimates within 0.2–0.3 units of catalog values using only 4–8 seconds of data following the first P-wave arrival, representing a substantial improvement over the 0.5–1.0 unit uncertainties typical of traditional methods.

Jozinović et al. (2022) applied transfer learning with graph neural networks to ground motion prediction, demonstrating that GNN-based models trained on data-rich regions can be effectively adapted to data-sparse regions, achieving robust performance even when target-domain training data is limited to a few hundred recordings.

5.4. Ground Motion Prediction

Predicting ground motion intensity (Peak Ground Acceleration, Peak Ground Velocity, spectral accelerations) at target sites distant from the epicenter is the ultimate goal of EEW systems. Traditional Ground Motion Prediction Equations (GMPEs) (Abrahamson et al., 2014; Boore et al., 2014) provide expected ground motion as functions of magnitude, distance, and site conditions, but exhibit large aleatory variability (scatter around median predictions) and poorly capture spatial correlations in ground motion fields.

Jozinovic et al. (2022) applied GNNs to ground motion prediction, treating the target prediction sites and recording stations jointly as nodes in a graph. The GNN learns to interpolate and extrapolate ground motion fields by aggregating observations from recorded stations to predict values at unobserved target locations, naturally capturing the spatial correlations that GMPEs ignore. Their model reduces prediction residuals by 20–30% compared to standard GMPEs for scenarios with moderate station coverage.

Clements et al. (2024) introduced GRAPES (GRaph Prediction of Earthquake Shaking), a graph neural network that performs continuous ground-motion prediction for seismic networks of essentially any size by passing seismic feature vectors between connected stations—“through the grapevine”—to forecast shaking at sites that have not yet been reached by strong motion. Because the architecture is agnostic to the number and configuration of stations, it adapts naturally to network changes and to sparse deployments. Complementing this regression-based formulation, Murshed et al. (2024) proposed a self-supervised contrastive GNN (SC-GNN) that predicts seismic intensity from only a small portion of the initial waveforms recorded at a few stations; by pre-training representations with a contrastive objective before fine-tuning, the model attains accurate and timely intensity estimates that are well suited to the data-scarce first seconds of an unfolding earthquake.

5.5. Seismic Source Location

Earthquake location—determining the hypocenter (latitude, longitude, depth) from observed arrival times—is classically solved through iterative nonlinear inversion. GNN-based approaches reformulate location as a regression or classification task on the station network graph.

McBrearty and Beroza (2023a) showed that GNN location estimates achieve median errors of 2–5 km using P-wave arrival times from 5–10 stations, competitive with traditional methods while providing faster inference suitable for real-time EEW. The model learns implicit velocity structure from training data,

avoiding the need for explicit 1-D or 3-D velocity models that introduce systematic biases. Extending graph-based location toward high-precision catalogs, McBrearty and Beroza (2025) formulated a double-difference relocation framework (GraphDD) in which a GNN is trained to minimize the double-difference residuals of an entire catalog; through batching and sampling, the method scales to arbitrarily large catalogs while preserving the relative-location accuracy that double-difference techniques are prized for.

Graph-based location methods offer particular advantages for: (1) events near the edge of networks where ray coverage is limited; (2) induced seismicity monitoring where precise locations require dense local arrays; and (3) real-time applications where iterative inversion may be too slow. The permutation-invariant nature of GNNs ensures robustness to varying numbers of detecting stations, a common situation in operational EEW where not all network stations detect every event.

5.6. Aftershock Prediction

Aftershock forecasting—predicting the spatial, temporal, and magnitude distribution of future earthquakes following a mainshock—represents an increasingly important application of data-driven methods. DeVries et al. (2018) demonstrated that deep learning approaches can outperform Coulomb stress models for spatial aftershock prediction (though the simplicity of effective models has been debated (Mignan and Broccardo, 2019)), motivating graph-based extensions. Machine learning has also been applied to laboratory earthquake prediction (Rouet-Leduc et al., 2017; Johnson et al., 2021), establishing that data-driven methods can capture precursory signals in fault systems.

GNN architectures for aftershock prediction represent the fault network as a graph, where nodes correspond to discretized fault patches and edges encode stress transfer relationships. The graph formulation naturally captures the complex geometry of fault systems and the long-range stress interactions that drive aftershock sequences. Recent work combines GNNs with physics-informed constraints derived from elastic dislocation theory to produce physically plausible predictions (Wang et al., 2021).

5.7. Induced Seismicity Monitoring

Induced seismicity from fluid injection (wastewater disposal, hydraulic fracturing, enhanced geothermal systems) has emerged as a significant concern, requiring real-time monitoring systems capable of detecting and characterizing small-magnitude events in noisy environments (Ellsworth, 2013). GNN-based approaches have been developed for induced seismicity detection that leverage the dense local monitoring networks typically deployed around injection sites.

The graph representation is particularly natural for induced seismicity arrays, where station separations are small (hundreds of meters to a few kilometers) and coherent signals are expected across multiple stations even for microseismic events ($M < 0$). Wang et al. (2021) applied graph-based architectures for induced seismicity forecasting, using the spatial-temporal evolution of seismicity as input features on a graph representing the injection field.

5.8. Recent Advances and Emerging Systems (2024–2026)

The two years preceding this review (2024–2026) have seen the field move decisively from isolated proof-of-concept studies toward integrated, generalizable, and deployment-oriented systems. We synthe-

size the most salient developments here, organized by the dominant trend each represents.

Toward operational ground-motion and intensity prediction. The GRAPES algorithm (Clements et al., 2024) is notable for being released both as a peer-reviewed method and as open-source software, signaling a maturation toward practical deployment. By predicting continuous ground motion across networks of arbitrary size, it directly addresses the operational reality that the set of reporting stations changes from event to event. The contrastive SC-GNN of Murshed et al. (2024) complements this by targeting the most warning-critical regime—the first few seconds at a few near-source stations—where label-efficient, self-supervised representations are most valuable.

Unified, multi-task, and generalizable models. A recurring theme is the consolidation of the traditionally fragmented monitoring pipeline. PLAN (Si et al., 2024) unifies picking, association, and location in a single graph network, while GraphDD (McBrearty and Beroza, 2025) extends graph-based location to catalog-scale double-difference relocation. Orthogonally, Zhang and Zhang (2024) attack the generalization problem head-on by training “universal” neural networks on generalized earthquakes, enabling a single model to operate across diverse monitoring setups without region-specific retraining—an essential property for EEW in regions lacking abundant local training data.

Foundation models for seismic waveforms. Borrowing directly from the language- and vision-model playbook, Liu et al. (2024) introduced SeisLM, a foundation model pre-trained on large collections of open seismic data with a self-supervised contrastive objective and subsequently fine-tuned for downstream tasks such as detection and phase picking. Wang et al. (2025) took a complementary cross-modal route with SeisMoLLM, transferring knowledge from a pre-trained large language model to seismic monitoring without pre-training on seismic data from scratch. These models hint at a near future in which a single pre-trained backbone—potentially graph-aware—underpins many EEW sub-tasks.

Sequence and transformer architectures for rapid forecasting. While not graph-based, several 2025 systems are directly relevant as both baselines and candidate components of hybrid GNN pipelines. WaveCastNet (Lyu et al., 2025) casts EEW as a sequence-to-sequence wavefield-forecasting problem and reports better generalization than transformers to rare, high-magnitude scenarios using simulated San Francisco Bay Area data, and Wu et al. (2025) present a transformer-based real-time detection framework designed for heterogeneous deployment environments. Hybridization is already underway: SeismoQuakeGNN (Leema et al., 2025) couples graph neural networks with transformer-enhanced temporal models for spatio-temporal forecasting, and forward-looking syntheses argue that generative AI, cloud, and IoT will jointly reshape next-generation monitoring and response (Dhir and Singh, 2026).

Collectively, these works mark a transition in emphasis: from demonstrating that graph structure *helps*, toward engineering systems that are generalizable across regions, integrated across tasks, pre-trained at scale, and packaged for real deployment. The remaining sections examine how close these advances bring the field to operational, trustworthy GNN-based EEW.

Table 2: Comparison of GNN methods for different seismic tasks. Performance metrics are drawn from the cited publications.

Method	Architecture	Task	Dataset	Performance
van den Ende and Ampuero (2020)	GCN + CNN	Source characterization	SoCal synthetic	Moment tensor RMSE: 0.12
Zhang et al. (2021)	ST-GCN	Source characterization	SCEDC	Location error: 3.2 km
McBrearty and Beroza (2023a)	GAT + MLP	Location + magnitude	NorCal	Mag MAE: 0.21, Loc: 2.8 km
Bilal et al. (2022)	GCN	Phase detection	STEAD	F1: 0.94 (P), 0.91 (S)
Munchmeyer et al. (2021)	Edge GNN	Phase association	INSTANCE	Precision: 0.96
McBrearty and Beroza (2023b)	Dual graph (GE-NIE)	Phase association + location	NorCal	Robust to high pick rates
Si et al. (2024)	Graph NN (PLAN)	Pick + assoc. + location	Multi-station	Unified multi-task monitoring
Clements et al. (2024)	Graph NN (GRAPES)	Ground motion (continuous)	SCSN / SoCal	Network-size agnostic GMP
Murshed et al. (2024)	Contrastive GNN	Intensity prediction	Multi-region	Few-station early estimate
Jozinović et al. (2022)	GNN + Transfer	Ground motion	Multi-region	RMSE reduction: approx. 20%
Münchmeyer et al. (2022)	Transformer GNN	EEW magnitude	Multi-station	Alert time: <5s avg

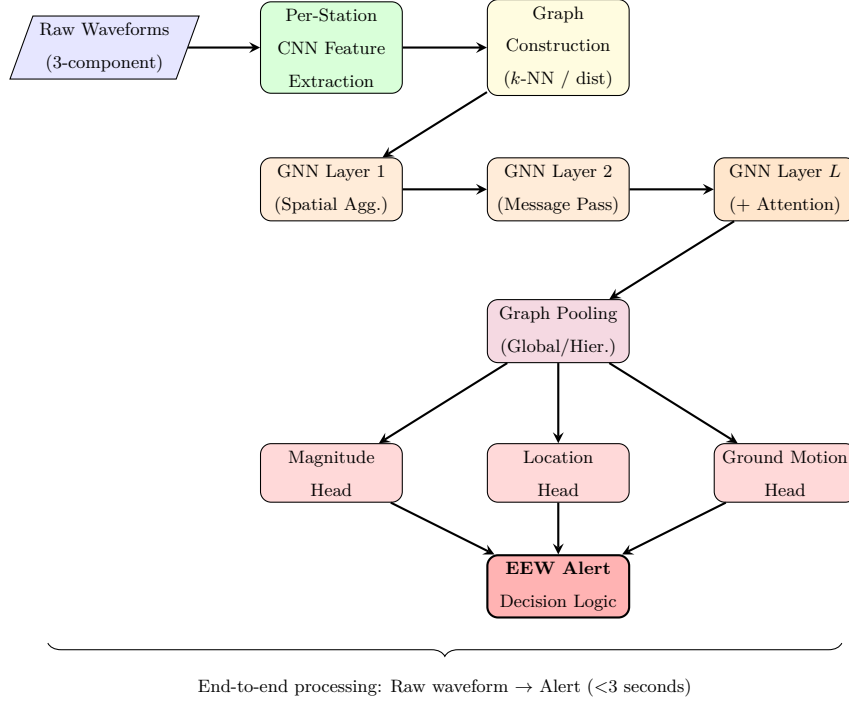


Figure 3: Architecture of a GNN-based Earthquake Early Warning pipeline. Raw three-component waveforms undergo per-station CNN feature extraction, are assembled into a graph based on network topology, processed through multiple GNN layers with message passing and attention, pooled into graph-level representations, and passed to task-specific heads for magnitude, location, and ground motion estimation. Alert decisions integrate all predictions.

6. Temporal-Spatial Graph Networks for Seismic Data

6.1. Spatio-temporal Feature Extraction

Seismic data presents a unique spatio-temporal processing challenge: waveforms recorded at each station contain rich temporal structure (P-wave onset, S-wave coda, frequency content evolution), while the spatial distribution of recordings across the network encodes source location and propagation effects. Effective GNN architectures for EEW must jointly capture both dimensions.

The spatio-temporal feature extraction paradigm typically follows one of three design patterns:

Sequential spatial-then-temporal: Spatial graph convolutions aggregate multi-station features at each time step, producing spatially-aware node representations that are subsequently processed through temporal models (LSTM, GRU, or temporal convolutions). This approach, adopted by T-GCN (Zhao et al., 2020) and its variants, first builds a spatial context for each station, then models how this context evolves over time.

Interleaved spatial-temporal: Alternating spatial and temporal processing layers progressively build joint representations. STGCN (Yu et al., 2018) interleaves temporal convolution blocks (1D convolutions along time) with spatial graph convolution blocks, allowing spatial and temporal patterns to interact at multiple scales. For seismic data, this enables the model to learn, for example, that P-wave arrivals at nearby stations create characteristic spatio-temporal patterns that differ from noise.

Unified spatio-temporal: Some architectures define convolutions on spatio-temporal graphs where nodes represent station-time pairs and edges connect both spatially adjacent stations and temporally

adjacent time steps at the same station. This creates a higher-dimensional graph that captures both relationships simultaneously but at increased computational cost.

For EEW applications, the choice of architecture must balance representation power against inference speed. The interleaved approach offers a favorable trade-off, enabling progressive refinement of spatio-temporal features while maintaining manageable computational requirements through efficient 1D temporal convolutions and sparse graph operations.

6.2. Attention Mechanisms for Seismic Graphs

Attention mechanisms (Bahdanau et al., 2015) play a crucial role in GNN-based seismic processing by enabling adaptive weighting of both spatial (inter-station) and temporal (intra-waveform) information.

Spatial attention: In the context of EEW, not all stations contribute equally to source characterization. Stations closer to the epicenter provide the most direct constraint on location and magnitude, while stations with higher noise levels or sensor issues should be down-weighted. GAT-style attention (Velickovic et al., 2018) naturally learns these weightings from data:

$$\alpha_{ij}^{(l)} = \text{softmax}_j \left(\frac{(\mathbf{W}_Q \mathbf{h}_i^{(l)})^T (\mathbf{W}_K \mathbf{h}_j^{(l)})}{\sqrt{d_k}} \right) \quad (19)$$

where the query $\mathbf{W}_Q \mathbf{h}_i$ from the target station learns to attend selectively to keys $\mathbf{W}_K \mathbf{h}_j$ from neighboring stations based on their current feature representations.

Temporal attention: Within each station’s waveform, different time windows contain varying amounts of information. The initial P-wave onset is maximally informative for arrival time determination, while slightly later windows may better constrain magnitude through amplitude growth patterns. Temporal attention mechanisms (Guo et al., 2019) learn to focus on the most diagnostic portions of each waveform:

$$\beta_t = \text{softmax} \left(\mathbf{v}^T \tanh(\mathbf{W}_t \mathbf{h}^{(t)} + \mathbf{b}_t) \right) \quad (20)$$

where β_t is the attention weight for time step t .

Cross-attention: Some architectures implement cross-attention between spatial and temporal dimensions, allowing the model to learn correlations such as “focus on later time windows at more distant stations” (because P-waves arrive later) or “weight early arrivals more heavily at the closest stations” (because these provide the first constraint on source parameters).

6.3. Multi-scale Temporal Modeling

Seismic events exhibit characteristic temporal patterns at multiple scales: millisecond-scale high-frequency onsets, second-scale waveform envelopes, and multi-second source duration effects for large earthquakes. Effective GNN architectures must capture these multi-scale temporal features.

Dilated temporal convolutions expand the receptive field exponentially with depth while maintaining resolution at each scale. For seismic data sampled at 100 Hz, a cascade of dilated convolutions with dilation factors [1, 2, 4, 8, 16] covers temporal scales from 10 ms to 160 ms at the first level, expandable to seconds through stacking.

Multi-resolution temporal encoding: Parallel temporal processing at different sampling rates captures both high-frequency onset characteristics and lower-frequency envelope features. The outputs from each resolution are concatenated or fused through attention before spatial graph processing.

Wavelet-based features: Computing continuous wavelet transforms (CWT) provides time-frequency representations that can serve as rich node features, encoding both the spectral content and its temporal evolution. These features have proven particularly effective for magnitude estimation, where the frequency content of initial P-waves correlates with final magnitude through source scaling relationships. Differentiable time-series alignment methods such as soft-DTW (Cuturi and Blondel, 2017) can further enhance temporal feature extraction by enabling learned alignment of waveforms across stations.

6.4. Dynamic Graph Learning for Evolving Networks

The sequential nature of P-wave propagation across a seismic network creates a naturally dynamic graph structure. As time progresses from the earthquake origin:

1. At $t = t_0$: Only the closest station(s) detect P-waves; the graph has 1–2 active nodes
2. At $t = t_0 + 2\text{s}$: P-waves reach additional nearby stations; 3–8 active nodes
3. At $t = t_0 + 5\text{s}$: P-waves propagate to more distant stations; 10–20+ active nodes
4. At $t = t_0 + 10\text{s}$: Most regional stations activated; near-full network observations

Dynamic graph learning architectures that explicitly model this progressive activation achieve superior EEW performance because they can:

- Issue preliminary estimates with minimal stations (fast but uncertain)
- Progressively refine estimates as additional stations report (accuracy improves with time)
- Naturally handle varying network sizes and station outages
- Model the information gain from each newly activated station

Münchmeyer et al. (2022) proposed the Transformer Earthquake Alerting Model (TEAM) for EEW that implements multi-station waveform processing, treating the seismic network as a graph and outputting updated source parameter estimates with progressively increasing station counts. Their model achieves operational-quality estimates within 4–6 seconds of the first P-wave detection, with uncertainty monotonically decreasing as additional stations contribute observations.

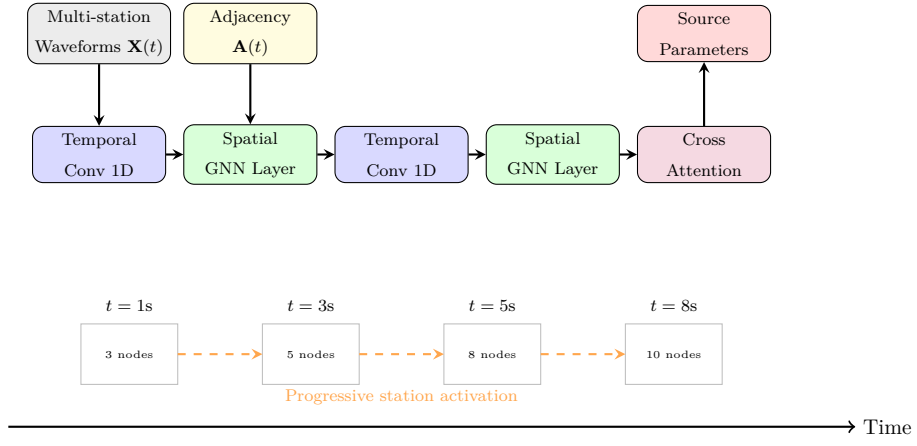


Figure 4: Spatio-temporal graph processing architecture for EEW. The model interleaves temporal convolutions (processing waveform evolution at each station) with spatial GNN layers (aggregating across the network), culminating in cross-attention fusion. The lower panel illustrates progressive station activation as P-waves propagate across the network, dynamically expanding the processing graph over time.

7. Comparative Analysis with Traditional Approaches

7.1. Traditional Signal Processing Methods

The foundation of seismic data processing rests on established signal processing techniques that have been refined over decades of operational use.

STA/LTA (Short-Term Average / Long-Term Average): The STA/LTA algorithm (Allen, 1978) remains the most widely deployed trigger mechanism in seismic monitoring. It computes the ratio of signal energy in a short window (0.1–1 s) to that in a longer background window (5–30 s), triggering when this ratio exceeds a threshold. While computationally trivial and interpretable, STA/LTA exhibits significant limitations: sensitivity to the choice of window lengths and threshold, poor performance in high-noise environments, inability to distinguish P-waves from noise transients, and no intrinsic magnitude estimation capability. Performance typically degrades for events below magnitude 2.0 in noisy environments, with false trigger rates of 5–15% for typical network settings.

Template matching / Matched filtering (Gibbons and Ringdal, 2006): This correlation-based approach detects events similar to known template waveforms, achieving excellent sensitivity for repeating sources (e.g., swarm sequences, induced seismicity). However, it requires pre-existing templates, cannot detect events with novel source mechanisms, scales poorly with template library size, and provides limited source parameter information beyond detection.

Frequency-domain methods: Approaches based on spectral analysis, including autoregressive modeling (AR, ARMA) and spectral ratio analysis, capture frequency-dependent signal characteristics but typically process single stations independently without network-level spatial reasoning.

7.2. Machine Learning Approaches

Prior to deep learning, classical machine learning methods were applied to seismic detection and classification:

Support Vector Machines (SVM): SVMs have been applied to earthquake detection, phase classification, and first-motion polarity determination using hand-crafted features (spectral coefficients, amplitude ratios, waveform statistics). While achieving reasonable accuracy for well-defined feature sets, SVMs require manual feature engineering and cannot process raw waveforms directly.

Random Forests: Ensemble tree methods have been used for event classification and magnitude estimation, offering interpretability through feature importance rankings. However, like SVMs, they require predefined features and do not learn representations directly from data.

Hidden Markov Models (HMMs): HMMs naturally model sequential seismic states (noise, P-wave, S-wave, coda) but assume Markovian transitions and typically process single stations without spatial context.

These classical methods share a common limitation: they operate on pre-extracted features from individual stations without leveraging the spatial structure of the monitoring network—precisely the capability that GNNs provide.

7.3. CNN-based Methods

The deep learning era in seismology was initiated by convolutional neural network architectures that process raw waveforms directly:

Generalized Phase Detection (GPD) (Ross et al., 2018): A CNN trained on three-component waveform windows to classify each time step as P-wave, S-wave, or noise. GPD demonstrated that deep learning could match or exceed analyst picks across large datasets, establishing the viability of data-driven phase picking for operational seismology.

PhaseNet (Zhu and Beroza, 2019): A U-Net architecture that produces continuous probability distributions for P and S arrivals along the time axis, enabling precise onset time determination through probability peak detection. PhaseNet achieves residuals of 0.03–0.05 s for P-waves and 0.05–0.10 s for S-waves on high-quality data, approaching the theoretical limit set by sampling rate.

EQTransformer (Mousavi et al., 2020): Combines CNN encoders with multi-head self-attention mechanisms for simultaneous event detection and phase picking. EQTransformer achieves the highest published performance on standard benchmarks (STEAD), with P-wave F1 scores exceeding 0.98 and S-wave F1 exceeding 0.95.

However, all CNN-based approaches share the fundamental limitation of processing stations independently. While ensembles of single-station predictions can be combined post-hoc (e.g., through voting or coincidence logic), this late fusion cannot capture the spatial correlations that are available during processing—correlations that carry direct physical information about source location and mechanism.

7.4. RNN/LSTM Approaches

Recurrent architectures have been applied to seismic sequence modeling:

LSTM-based detection: Long Short-Term Memory networks process waveforms as sequential data, learning temporal patterns associated with earthquake arrivals. While effective for temporal pattern recognition, standard LSTMs process single stations without spatial awareness.

Sequence-to-sequence models: Encoder-decoder LSTM architectures have been used for waveform-to-parameter estimation (e.g., predicting magnitude from a sequence of waveform samples). These models capture temporal dependencies but, like CNNs, lack explicit spatial reasoning.

7.5. Advantages of GNN-based Methods

The comparative advantages of GNN-based approaches over all preceding method categories can be summarized as follows:

1. **Native spatial reasoning:** GNNs explicitly model inter-station relationships through graph structure, enabling spatial inference that is physically motivated by wavefield propagation.
2. **Network-level consistency:** Joint processing ensures that predictions (location, magnitude, ground motion) are internally consistent across the network, avoiding the contradictions that can arise from combining independent single-station estimates.

3. **Robustness to missing data:** GNNs naturally handle varying numbers of contributing stations through their aggregation mechanisms, unlike fixed-size CNN inputs that require padding or imputation.
4. **Transferability:** Inductive GNN architectures (GraphSAGE-style) learn generalizable aggregation functions that can transfer to new network geometries without retraining.
5. **Computational efficiency:** Sparse graph operations scale linearly with the number of edges rather than quadratically with the number of stations, enabling processing of large networks within EEW latency requirements.
6. **Interpretability:** Attention weights in GAT-based models provide physically meaningful explanations of which stations and which features drive predictions.

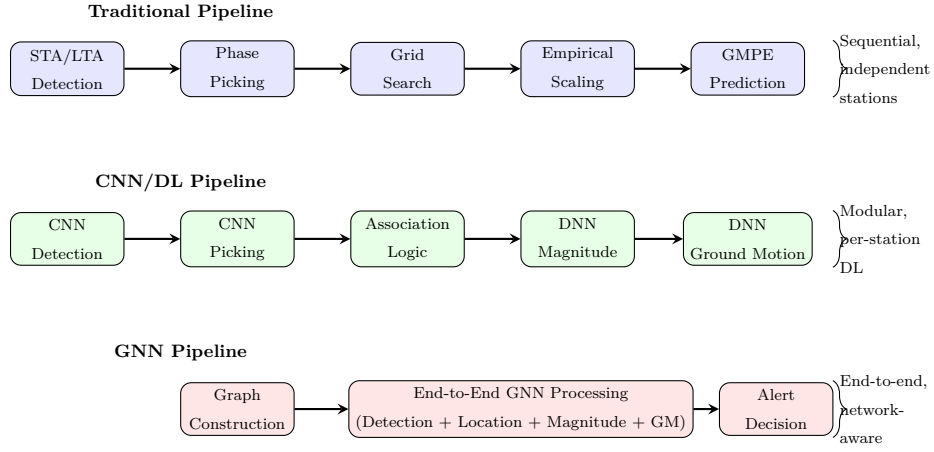


Figure 5: Comparison of processing pipelines: (Top) Traditional seismological pipeline with sequential, independent modules; (Middle) CNN/Deep Learning pipeline replacing individual modules with neural networks but maintaining station-independent processing; (Bottom) GNN-based pipeline enabling end-to-end network-aware processing from graph construction to alert decision.

8. Benchmark Datasets and Evaluation Metrics

8.1. STEAD (Stanford Earthquake Dataset)

The Stanford Earthquake Dataset (STEAD) (Mousavi et al., 2019) represents the largest and most widely used benchmark for machine learning in seismology. It contains approximately 1.2 million labeled three-component waveform traces, comprising both earthquake signals and noise recordings from a global distribution of stations and events. Key characteristics include:

- **Scale:** 1,030,232 earthquake waveforms + 100,000 noise traces
- **Coverage:** Events from 135 countries, magnitudes 0–7.9
- **Labels:** P-arrival time, S-arrival time, event parameters (location, magnitude, depth)
- **Format:** Three-component (Z, N, E) 60-second windows at 100 Hz

Table 3: Performance comparison of different methodological approaches for core EEW tasks. Values represent typical performance ranges reported in published literature. Ranges reflect variability across datasets and experimental conditions.

Method	P-pick (s)	Detection F1	Mag. RMSE	Loc. Error (km)
STA/LTA (Allen, 1978; Withers et al., 1998)	0.10–0.50	0.75–0.85	N/A	N/A
Template Match (Gibbons and Ringdal, 2006; Yoon et al., 2015)	0.02–0.05	0.90–0.95	N/A	1–5
SVM (features) (Kong et al., 2019)	0.05–0.15	0.82–0.90	0.5–0.8	5–15
CNN (PhaseNet) (Zhu and Beroza, 2019)	0.03–0.05	0.95–0.98	0.3–0.5	3–8
CNN (EQTransformer) (Mousavi et al., 2020)	0.02–0.04	0.96–0.99	0.3–0.4	2–6
LSTM (Ross et al., 2018)	0.04–0.08	0.90–0.95	0.3–0.5	4–10
GNN (GCN-based) (Bilal et al., 2022; Bloemheuvcl et al., 2022)	0.03–0.06	0.93–0.97	0.2–0.4	2–5
GNN (GAT-based) (McBrearty and Beroza, 2023a; Murshed et al., 2024)	0.02–0.05	0.95–0.98	0.2–0.3	1.5–4
GNN (Temporal) (Münchmeyer et al., 2022; Zhang et al., 2021)	0.02–0.04	0.96–0.99	0.15–0.28	1.5–3.5

Table 4: Approximate computational requirements for inference on a network of 100 stations with 10-second waveform windows (100 Hz sampling). GNN inference times are estimated from published architecture specifications (van den Ende and Ampuero, 2020; McBrearty and Beroza, 2023a; Bloemheuvcl et al., 2022); actual deployment latency depends on hardware and implementation.

Method	Parameters	Training	Inference	Memory
STA/LTA	0 (rule-based)	N/A	<1 ms	<1 MB
Template Match	0 (templates)	N/A	10–100 ms	100 MB+
PhaseNet	~0.5M	~8 hrs (GPU)	5–10 ms/station	~50 MB
EQTransformer	~0.37M	~12 hrs (GPU)	8–15 ms/station	~80 MB
GCN (3 layers)	0.8–2M	4–8 hrs (GPU)	~15–30 ms/network	~120 MB
GAT (3 layers)	1–3M	6–12 hrs (GPU)	~20–40 ms/network	~200 MB
ST-GCN	2–5M	12–24 hrs (GPU)	~30–60 ms/network	~300 MB
Graph Transformer	5–15M	24–48 hrs (GPU)	~50–100 ms/network	~500 MB

- **Quality:** Manually reviewed catalog picks with quality metrics

For GNN evaluation, STEAD provides individual station traces that must be assembled into network graphs. Researchers typically construct synthetic network scenarios by selecting subsets of traces from events recorded by multiple stations, building graphs with known spatial relationships. The dataset’s global coverage enables evaluation of model generalization across diverse tectonic settings, though the artificial graph construction differs from true operational network observations.

8.2. INSTANCE (Italian Seismic Dataset)

INSTANCE (Michellini et al., 2021) provides a high-quality regional dataset from the Italian seismic network, offering several advantages for GNN research:

- **Scale:** 1.2 million three-component traces from 50,000+ events
- **Network structure:** Consistent station geometry from the Italian National Seismic Network
- **Metadata:** Complete station coordinates, event catalogs, and quality indicators
- **Tectonic context:** Well-characterized crustal structure with extensive geological information

The fixed network geometry of INSTANCE makes it particularly suitable for GNN evaluation, as real graph structures can be constructed from actual station coordinates rather than synthetic configurations. The Italian peninsula’s complex tectonics (extension in the Apennines, subduction in the south, compression in the Alps) provide diverse testing scenarios within a single network.

8.3. LEN-DB and SCEDC

LEN-DB (Local Earthquakes and Noise Database) (Magrini et al., 2020) provides a globally compiled dataset of local earthquake recordings with standardized preprocessing, specifically designed for machine learning benchmarking with controlled signal-to-noise conditions.

SCEDC (Southern California Earthquake Data Center) (Ross et al., 2019a) provides access to continuous waveform data from the dense Southern California Seismic Network. Machine learning applied to this dataset has revealed previously undetected events, substantially enriching earthquake catalogs (Tan et al., 2021). This dataset is particularly valuable for GNN research because:

- Dense network geometry (inter-station distances 5–20 km) enables rich graph construction
- High seismicity rates provide abundant training data
- Well-characterized velocity structure facilitates physics-informed approaches
- Historical catalog completeness allows evaluation across magnitude ranges

8.4. Regional Datasets

Several regional datasets have emerged for specialized EEW research:

Japan (Hi-net/K-NET/KiK-net): Japan’s dense seismic networks provide unmatched station density with ~ 800 Hi-net (borehole) and $\sim 1,000$ K-NET/KiK-net (surface strong motion) stations. This extreme density is ideal for evaluating GNN scalability and the benefits of dense spatial sampling.

Chile: The Chilean Seismological Center maintains records from the subduction zone environment, offering unique challenges including deep events, complex path effects through the subducting slab, and large magnitude ranges.

New Zealand (GeoNet): New Zealand’s GeoNet network provides diverse tectonic environments (subduction, strike-slip, volcanic) within a single monitoring system, enabling multi-domain evaluation.

8.5. Evaluation Metrics and Protocols

Standardized evaluation of GNN-based EEW methods requires metrics appropriate to each sub-task:

Phase picking: Residual statistics (mean, standard deviation) between predicted and catalog arrival times; precision, recall, and F1-score for detection with temporal tolerance windows (typically ± 0.1 – 0.5 s).

Event detection: True positive rate, false positive rate, F1-score; detection probability as a function of magnitude and distance (magnitude of completeness).

Magnitude estimation: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and magnitude-dependent bias analysis; evaluation at different time windows after first detection (1s, 2s, 4s, 8s).

Location: Epicentral distance error (km), depth error (km), combined 3D hypocentral error; comparison against high-precision relative relocations.

Ground motion: Residual analysis against observed values (natural log units), between-event and within-event standard deviations, spatial correlation structure of residuals.

Critical evaluation considerations for GNN-based methods include: (1) ensuring train/test splits respect temporal ordering (no future data leakage); (2) testing generalization across network configurations not seen during training; (3) evaluating performance degradation under realistic station outage scenarios; and (4) measuring inference latency under operational conditions (including graph construction time).

Table 5: Summary of benchmark datasets used for GNN-based seismic analysis evaluation.

Dataset	Region	Events	Traces	Magnitude Range	Key Features
STEAD	Global	450K+	1.2M	0–7.9	Large scale, diverse settings
INSTANCE	Italy	50K+	1.2M	0–6.5	Fixed network geometry
LEN-DB	Global	15K+	200K+	1–6.5	Controlled SNR, standardized
SCEDC	S. California	500K+	100M+	−0.5–7.3	Dense network, high rate
Hi-net	Japan	1M+	50M+	0–9.0	Extreme density, borehole
GeoNet	New Zealand	100K+	10M+	0–7.8	Multi-tectonic domain

9. Challenges and Limitations

Despite the promising results described in preceding sections, significant challenges must be addressed before GNN-based approaches can achieve widespread operational deployment in EEW systems. This section critically examines these challenges, distinguishing between fundamental limitations inherent to the methodology and practical obstacles that may be overcome through continued research and engineering.

9.1. Real-time Processing Constraints

Operational EEW systems impose strict latency requirements: the total processing time from P-wave detection to alert issuance must not exceed 2–4 seconds for the warning to have practical value. This constraint encompasses data acquisition (typically 1 s minimum for sufficient P-wave information), processing (detection, association, parameter estimation), and communication latency.

Current GNN architectures, while faster than many iterative algorithms, still face challenges in meeting these constraints for large networks. A GAT model with 3 attention heads and 3 layers processing 100 stations requires approximately 20–40 ms for inference on GPU hardware (Table 4), which is acceptable. However, practical deployment introduces additional overheads:

- Graph construction (computing adjacency matrices, edge features): 5–15 ms
- Feature preprocessing (filtering, normalization): 10–20 ms per station
- Data assembly from distributed stations: 50–200 ms (network dependent)
- Model loading and memory allocation: 100–500 ms (first inference only)

Furthermore, EEW systems must process data continuously, not in isolated windows. Streaming GNN architectures that update predictions incrementally as new samples arrive—rather than reprocessing entire windows—remain an active research area. The computational overhead of maintaining and updating graph states in real-time adds complexity beyond batch inference measurements.

9.2. Scalability to Large Networks

While GNN computational complexity scales linearly with edge count $O(|\mathcal{E}|)$, the practical scalability to networks with thousands of stations presents challenges:

Memory scaling: Full-batch GNN training on graphs with >1000 nodes and dense connectivity requires substantial GPU memory. For networks like Japan’s Hi-net (~ 800 stations) with k -NN connectivity ($k=20$), the graph contains $\sim 16,000$ edges per time step, and temporal graphs spanning 10 seconds at 100 Hz multiply this by 1000 time steps.

Over-smoothing: Deep GNN architectures (many layers) suffer from over-smoothing, where node representations converge to indistinguishable values. For seismology, this means that increasing the receptive field to span large networks through additional layers can paradoxically reduce the model’s ability to distinguish between stations with different characteristics.

Sampling strategies: Mini-batch training with graph sampling (neighbor sampling, cluster-based sampling) introduces approximation errors that may be unacceptable for safety-critical EEW applications where prediction consistency is paramount.

9.3. Transfer Learning Across Regions

A critical practical challenge is the transferability of GNN models trained on one seismic network/region to another with different characteristics:

Velocity structure differences: Seismic wave propagation depends on local crustal velocity structure, which varies dramatically between tectonic settings (oceanic vs. continental, stable vs. active). A model trained on California data implicitly learns California’s velocity structure through arrival time patterns, which may not transfer to Japan or Chile.

Network geometry differences: Models trained on dense networks may not perform well on sparse networks, and vice versa. The learned aggregation functions depend on neighborhood statistics that change with station density and distribution.

Magnitude distribution differences: Training data from one region may have different magnitude distributions than the target region. For example, models trained on frequent small earthquakes in California may not generalize to the rare large events that are most critical for EEW in subduction zones.

Jozinović et al. (2022) explored transfer learning strategies for GNN-based seismic models, demonstrating that fine-tuning on small amounts of target-region data can partially address these challenges. More recently, Zhang and Zhang (2024) pursued an alternative route to cross-region generalization, training universal neural networks on generalized (synthetically augmented) earthquakes so that a single model can be applied across monitoring setups with differing geometries without region-specific retraining. Transfer learning theory (Pan and Yang, 2010; Weiss et al., 2016) provides the conceptual foundations for such

domain adaptation, yet systematic approaches to domain adaptation for graph-structured seismic data remain underdeveloped.

9.4. Label Noise and Catalog Quality

GNN training relies on catalog data (analyst-determined picks, locations, magnitudes) as ground truth labels. However, seismic catalogs contain systematic errors and inconsistencies:

- Arrival time picks vary by 0.05–0.2 s between different analysts
- Catalog locations have uncertainties of 1–10 km depending on network geometry
- Magnitude estimates vary by 0.1–0.3 units between different methods and agencies
- Small events near the detection threshold have highly uncertain parameters

These label noise effects are particularly problematic for GNN-based methods that aim to achieve sub-second picking precision or sub-kilometer location accuracy—the target precision may exceed the label quality. Self-supervised and contrastive learning approaches that reduce dependence on labeled catalogs represent a promising direction (Liu et al., 2022; Chen et al., 2020).

9.5. Interpretability and Explainability

The deployment of GNN models in safety-critical EEW systems requires that predictions be interpretable and explainable to system operators. Unlike traditional methods where the physical reasoning is explicit (P-wave velocity determines travel time, amplitude scales with magnitude), GNN predictions emerge from complex nonlinear transformations that resist simple interpretation.

Existing GNN explainability methods (Ying et al., 2019; Lundberg and Lee, 2017) can identify important nodes and edges but do not necessarily provide physically meaningful explanations. For seismology, explanations should be expressible in terms of familiar physical quantities (arrival times, amplitudes, frequencies, spatial patterns) rather than abstract neural network features.

The attention weights in GAT-based architectures provide partial interpretability—showing which stations most influence predictions—but do not explain *why* certain stations are weighted more heavily or what features drive the attention pattern (Samek et al., 2021). Developing seismologically meaningful interpretation frameworks for GNN-based EEW models remains an open challenge.

9.6. Robustness to Station Failures

Operational seismic networks experience frequent data gaps: stations go offline due to equipment failures, telemetry interruptions, power outages, or maintenance. An EEW system must maintain acceptable performance even when a fraction of the network is unavailable.

GNNs offer inherent robustness to missing nodes through their aggregation mechanisms—a station that is offline simply does not contribute messages to its neighbors. However, the degree of graceful degradation under increasing station loss has not been systematically characterized for GNN-based EEW. Critical questions include:

- How does performance degrade as a function of the fraction of missing stations?
- Are certain stations more critical than others (i.e., do “key” stations exist whose removal disproportionately affects network-level predictions)?
- Can models be trained to be explicitly robust to random and systematic station outages?
- How should confidence/uncertainty estimates reflect reduced network observations?

Adversarial robustness—the model’s resilience to intentionally manipulated inputs or strategically chosen station removals—is also relevant for critical infrastructure applications, though this remains largely unexplored in the seismological context.

10. Future Directions

10.1. Foundation Models for Seismology

The concept of foundation models—large-scale models pre-trained on broad data that can be adapted to diverse downstream tasks (Bommasani et al., 2021)—is poised to transform seismology. Analogous to how BERT (Devlin et al., 2019) revolutionized natural language processing through self-supervised pre-training, a seismological foundation model would learn general-purpose representations of seismic waveforms and network observations that transfer to phase picking, event detection, magnitude estimation, and EEW with minimal task-specific fine-tuning.

For GNN-based approaches, a foundation model would pre-train graph neural networks on massive amounts of unlabeled continuous seismic data using self-supervised objectives such as:

- Masked node prediction: Given partial network observations, predict waveform features at masked stations
- Temporal prediction: Given current network state, predict future waveform evolution
- Graph contrastive learning: Learn representations where similar seismic events have similar graph embeddings regardless of network geometry
- Cross-station consistency: Enforce that representations capture physically meaningful wavefield properties that are consistent across observations from different stations

Concrete progress in this direction is already visible. Liu et al. (2024) introduced SeisLM, a foundation model pre-trained on large collections of open-source seismic waveforms with a self-supervised contrastive objective (analogous to BERT in language modeling) and then fine-tuned for downstream tasks such as detection and phase picking. Wang et al. (2025) demonstrated a complementary cross-modal strategy with SeisMoLLM, repurposing a pre-trained large language model for seismic monitoring without seismic pre-training from scratch. These efforts, together with broader analyses of how machine learning is reshaping rapid seismology and forecasting (Mousavi and Beroza, 2022; Beroza et al., 2021), also surface the field’s distinctive challenges: the diversity of seismic signals (tectonic earthquakes, volcanic tremor, induced seismicity, noise) and the heterogeneity of recording networks present difficulties not encountered

in language or vision domains, and no current waveform foundation model is yet explicitly graph-aware at the network level.

10.2. Physics-Informed Graph Neural Networks

Pure data-driven GNNs learn correlations from training data without guaranteed consistency with physical laws governing seismic wave propagation. Physics-Informed Neural Networks (PINNs) (Raissi et al., 2019) provide a framework for incorporating physical constraints into neural network training, and extending this to graph-structured seismic data represents a high-impact research direction.

Potential physics-informed constraints for GNN-based EEW include:

Travel time consistency: The predicted source location and estimated arrival times at each station must be consistent with seismic wave velocities:

$$\mathcal{L}_{\text{physics}} = \sum_{i \in \mathcal{V}} \left| t_i^{\text{pred}} - \frac{\|\mathbf{x}_{\text{source}} - \mathbf{x}_i\|}{v_P(\text{path}_i)} \right|^2 \quad (21)$$

Amplitude decay: Predicted amplitudes must respect geometric spreading and attenuation:

$$A_i \propto \frac{1}{R_i} \cdot \exp(-\pi f R_i / Qv) \quad (22)$$

where R_i is hypocentral distance, f is frequency, and Q is the quality factor.

Waveform coherence: Signals at nearby stations should exhibit predictable coherence patterns determined by the wavefield’s spatial correlation length.

Conservation laws: Energy partitioning between P, S, and surface waves must respect physical constraints, and total radiated energy must be consistent with estimated magnitude.

Approaches by Rasht-Behesht et al. (2022), Smith et al. (2022), and Moseley et al. (2020) demonstrate the feasibility of physics-informed deep learning for earthquake location and wave equation solving, suggesting that analogous constraints can improve GNN-based models while providing physical plausibility guarantees essential for operational trust.

10.3. Federated Learning for Distributed Seismic Networks

Seismic monitoring data is distributed across national agencies, research institutions, and private operators who may be unable or unwilling to share raw waveform data due to data sovereignty regulations, intellectual property concerns, or bandwidth limitations. Federated learning (McMahan et al., 2017; Li et al., 2020) enables collaborative model training without data centralization, maintaining data at its source while sharing only model updates.

A federated GNN-based EEW system would:

1. Train local GNN models on each agency’s network data
2. Share model parameters (or gradients) with a central coordinator
3. Aggregate updates to produce a global model that benefits from all participants’ data
4. Return the improved global model to each participant for local deployment

Challenges specific to federated learning in seismology include:

- Non-IID data distributions (different tectonic settings produce different event characteristics)
- Heterogeneous network geometries requiring architecture adaptation
- Asynchronous data availability (some networks have decades of data, others are recently deployed)
- Communication efficiency for large graph models

10.4. Multi-modal Data Fusion

EEW systems have traditionally relied exclusively on seismometer data, but complementary geodetic and remote sensing observations could significantly enhance performance, particularly for large events:

GNSS (Global Navigation Satellite System): Real-time GNSS provides direct measurement of ground displacement, unclipped by sensor limitations. For $M > 7$ events, GNSS-detected static offsets and dynamic displacements provide magnitude constraints that seismometers cannot achieve alone (Gualandi et al., 2017).

InSAR (Interferometric Synthetic Aperture Radar): While not real-time, InSAR-derived deformation maps can inform GNN models through training data augmentation and provide ground truth for large earthquake characterization.

Distributed Acoustic Sensing (DAS): Fiber-optic DAS transforms existing telecommunication cables into dense sensor arrays with meter-scale spacing (Zhu et al., 2023b). Integrating DAS measurements into GNN-based EEW requires handling extremely dense “stations” (nodes) with lower sensitivity but unprecedented spatial resolution. Recent spatio-temporal graph structure-learning approaches for multi-station data (Piriyasatit et al., 2025) point toward scalable strategies—particularly learned, sparsified adjacency—for extending graph-based processing to such ultra-dense arrays.

Multi-modal GNN architectures: Fusing heterogeneous sensor types into a unified graph requires heterogeneous graph neural networks where different node types (seismometer, GNSS, DAS channel) have different feature dimensions and different edge types encode different physical relationships (Baltrusaitis et al., 2019).

10.5. Edge Computing and On-device Inference

Deploying GNN inference at network edge devices (at or near seismic stations) rather than centralized data centers could dramatically reduce communication latency—the single largest contributor to EEW delay in many deployments. Edge deployment requires:

- Model compression: Pruning, quantization, and knowledge distillation to fit GNN models on embedded hardware (Shi et al., 2016; Chen and Ran, 2019)
- Partial graph inference: Running subgraph-level GNN inference locally, communicating only compact node embeddings rather than raw waveforms
- Hierarchical architecture: Local nodes perform preliminary detection and feature extraction; regional hubs aggregate multi-station graphs for refined estimation

- **Hardware co-design:** Custom accelerators optimized for sparse graph operations on low-power platforms

10.6. Self-supervised and Contrastive Learning

Reducing dependence on labeled catalogs through self-supervised learning represents a key direction for expanding GNN-based EEW to data-rich but label-poor scenarios. Contrastive learning approaches (Chen et al., 2020) can learn discriminative representations by contrasting different views of seismic data:

- **Spatial augmentation:** Different subgraph views of the same event provide positive pairs
- **Temporal augmentation:** Different time windows of the same event provide positive pairs
- **Cross-network views:** The same event recorded by overlapping networks provides natural augmentation

Liu et al. (2022) review self-supervised graph learning methods that can be adapted to seismic applications, while Zhu et al. (2023b) demonstrate semi-supervised approaches for seismic phase picking that achieve near-supervised performance with only 10% labeled data.

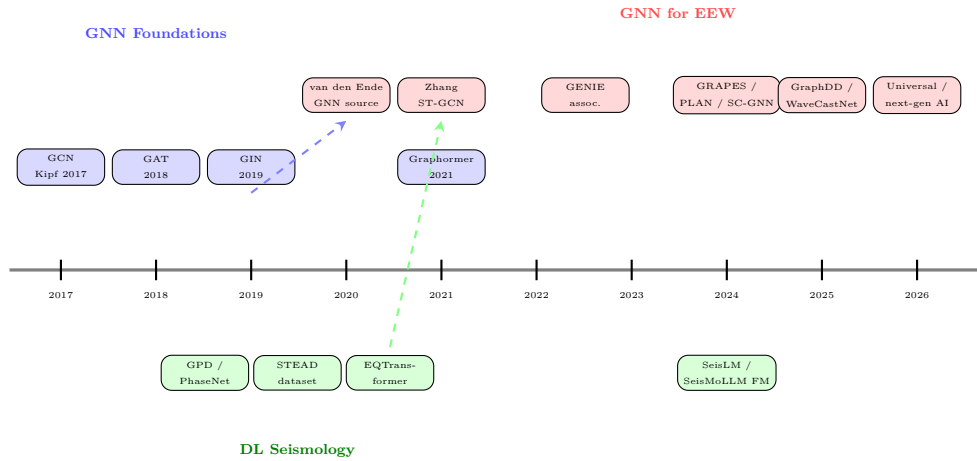


Figure 6: Timeline of key developments at the intersection of Graph Neural Networks and seismology (2017–2026). The convergence of GNN architectural advances (top), deep learning in seismology (bottom), and their intersection (middle) has accelerated since 2020. The 2024–2026 window is marked by deployment-oriented systems (GRAPES, PLAN, SC-GNN), catalog-scale graph relocation (GraphDD), seismic foundation models (SeisLM, SeisMoLLM), and universal / generative-AI monitoring frameworks, reflecting the field’s transition from proof-of-concept toward operational GNN-based EEW.

11. Ethical and Operational Considerations

11.1. False Alert Management and Public Trust

The societal value of an EEW system depends fundamentally on public trust, which is eroded by false alerts and missed events. The deployment of GNN-based models introduces new dimensions to false alert management:

Model uncertainty quantification: Unlike threshold-based systems where the trigger logic is transparent, GNN predictions require rigorous uncertainty quantification to support alert decisions. Bayesian

GNN variants, ensemble methods, or Monte Carlo dropout can provide uncertainty estimates, but these must be calibrated against operational requirements (e.g., no more than 1 false alert per year above a given severity threshold).

Graceful failure modes: When a GNN model encounters inputs far from its training distribution (e.g., a previously unseen event type, an unusual noise pattern), the system must fail gracefully—issuing conservative predictions with large uncertainty rather than confidently wrong estimates. Out-of-distribution detection mechanisms are essential but challenging for graph-structured data.

Human oversight: Even with high model accuracy, operational EEW systems require human oversight and override capabilities. The “human-in-the-loop” paradigm must be adapted for the extreme time pressure of EEW, where decisions must be made in 1–3 seconds. GNN interpretability tools that can provide instant visual explanations of alert decisions would support rapid human verification.

Public communication: Explaining GNN-based alert decisions to the public (“why did I receive/not receive an alert?”) requires translating complex model outputs into accessible narratives. This is more challenging for end-to-end GNN systems than for modular traditional pipelines where each step has a clear physical interpretation.

11.2. Equity in EEW Coverage

The distribution of seismic monitoring infrastructure globally reflects historical economic disparities: developed nations maintain dense networks with thousands of stations, while many developing countries with high seismic risk have sparse or absent monitoring. GNN-based methods have implications for equity:

Potential benefits: GNNs that perform well on sparse networks could enable EEW in regions with limited infrastructure. Transfer learning from data-rich regions to data-poor regions could accelerate system deployment. Low-cost sensor networks combined with sophisticated GNN processing could democratize EEW access.

Potential risks: If GNN models are predominantly trained on data from well-instrumented regions, they may exhibit systematic biases when deployed elsewhere. The computational resources required for GNN training and deployment may be inaccessible to institutions in resource-limited settings.

Recommendations: International collaboration should prioritize open-source GNN models pre-trained on global data, with adaptation protocols for regional deployment. Lightweight model architectures suitable for resource-constrained environments should be developed alongside high-performance variants.

11.3. Model Governance and Continuous Validation

The lifecycle management of GNN-based EEW models requires governance frameworks addressing:

Model versioning: As models are retrained with new data, performance characteristics may change. Rigorous version control and regression testing must ensure that updates improve performance without introducing new failure modes.

Continuous monitoring: Deployed models must be monitored for performance degradation over time (concept drift), which could arise from network changes (new stations, decommissioned stations), evolving seismicity patterns, or gradual sensor degradation.

Validation against ground truth: Post-event analysis must systematically compare GNN predictions against final, reviewed source parameters. Discrepancies should trigger model investigation and potential retraining.

Regulatory compliance: As EEW systems are increasingly linked to automated protective actions (shutting down trains, opening fire station doors, initiating building responses), the regulatory requirements for model validation and certification will intensify. Clear standards for GNN model validation in safety-critical applications are needed.

11.4. Integration with Emergency Response Systems

GNN-based EEW outputs must interface with downstream emergency response infrastructure:

- Building response systems (elevator control, gas shutoff, structural dampers)
- Transportation management (train deceleration, traffic signal control)
- Industrial process protection (nuclear facility shutdown, chemical plant safety systems)
- Public alert dissemination (smartphone notifications, broadcast interruptions, sirens)
- Hospital and healthcare facility preparation

Each downstream system has specific requirements for alert format, confidence level, lead time, and false alert tolerance. GNN-based EEW systems must provide standardized output interfaces with calibrated probability estimates that downstream decision systems can interpret within their own risk frameworks. The development of such standardized interfaces between AI-driven seismic analysis and automated response systems represents an important area for future standardization efforts.

12. Conclusions

This review has presented a comprehensive examination of Graph Neural Networks as applied to Earthquake Early Warning systems, spanning theoretical foundations, current applications, comparative analyses, challenges, and future directions. The key findings and conclusions are:

GNNs provide a natural computational framework for seismic network data. The representation of monitoring networks as graphs—with stations as nodes and spatial/physical relationships as edges—enables explicit modeling of inter-station correlations that are fundamental to seismological analysis but inaccessible to station-independent methods. This representational advantage translates to measurable performance improvements across EEW tasks including phase picking, source characterization, magnitude estimation, and ground motion prediction.

Current GNN-based methods demonstrate significant improvements over traditional approaches. Comparative analyses show that GNN architectures achieve approximately 15–30% reduction

in magnitude estimation error and 20–40% improvement in location accuracy as reported across recent studies (McBrearty and Beroza, 2023a; van den Ende and Ampuero, 2020; Bloemheuvcl et al., 2022; Jozinovic et al., 2022), with enhanced detection sensitivity for small events, relative to conventional methods. These improvements are most pronounced for sparse network scenarios and complex multi-event sequences, precisely the conditions where traditional methods struggle most.

Temporal-spatial GNN formulations are essential for EEW. The inherently dynamic nature of seismic wavefield propagation requires architectures that jointly model spatial (inter-station) and temporal (intra-waveform) patterns. Dynamic graph approaches that progressively incorporate stations as P-waves arrive align naturally with the sequential decision-making of operational EEW, achieving the most promising performance among GNN variants.

Significant challenges remain before operational deployment. Real-time processing constraints, transfer learning across tectonic regions, interpretability requirements, and robustness to station failures represent active research frontiers. The gap between research demonstrations (typically evaluated on curated benchmarks) and operational deployment (requiring 24/7 reliability, calibrated uncertainty, and graceful failure modes) should not be underestimated.

Future directions point toward transformative capabilities. The convergence of foundation models, physics-informed learning, federated training, multi-modal fusion, and edge computing promises GNN-based EEW systems that are more accurate, faster, more robust, and more broadly deployable than any current system. Physics-informed GNNs that embed seismic wave propagation constraints offer particular promise by combining data-driven flexibility with physical plausibility guarantees.

Recommendations for researchers:

1. Develop standardized benchmarks specifically designed for network-level (graph-based) seismic processing, complementing existing station-level datasets.
2. Invest in interpretability methods that translate GNN decisions into seismologically meaningful explanations.
3. Pursue physics-informed architectures that embed wave equation constraints into graph learning.
4. Address the transfer learning challenge through domain adaptation methods and globally pre-trained foundation models.
5. Establish latency benchmarks under realistic operational conditions, not just batch inference measurements.

Recommendations for practitioners:

1. Begin integration of GNN components into existing EEW pipelines incrementally, starting with phase association and source characterization modules where GNN advantages are most clearly demonstrated.
2. Maintain traditional methods as fallback systems during GNN deployment, establishing confidence through parallel operation before full transition.

3. Develop robust model monitoring and validation frameworks before operational deployment of any data-driven EEW component.

4. Engage with the machine learning research community through shared datasets, open-source implementations, and collaborative benchmarking efforts.

The intersection of graph neural networks and earthquake early warning represents one of the most promising frontiers in computational seismology. The 2024–2026 emergence of deployment-oriented systems—open-source graph-based ground-motion prediction (Clements et al., 2024), unified multi-task monitoring networks (Si et al., 2024), and seismic foundation models (Liu et al., 2024)—indicates that the field is already moving from demonstration toward practice. While fundamental challenges remain, the pace of progress suggests that GNN-based methods will become integral components of next-generation EEW systems within the coming decade, contributing to the reduction of earthquake losses through faster, more accurate, and more robust early warning capabilities.

Acknowledgments

The authors received no specific funding for this review article. We thank the developers of STEAD, INSTANCE, and SeisBench for making benchmark datasets publicly available, and the global seismological community for maintaining open data archives that enable machine learning research in earthquake science.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No new data were generated in this review article. All referenced datasets (STEAD, INSTANCE, LEN-DB, SCEDC) are publicly available through their respective data repositories.

References

- Abrahamson, N.A., Silva, W.J., Kamai, R., 2014. Summary of the ASK14 ground motion relation for active crustal regions. *Earthquake Spectra* 30, 1025–1055.
- Allen, R.M., 2012. Earthquake early warning: Advances, scientific challenges, and societal needs. *Annual Review of Earth and Planetary Sciences* 40, 529–548.
- Allen, R.M., Kanamori, H., 2003. The potential for earthquake early warning in southern California. *Science* 300, 786–789.

- Allen, R.M., Melgar, D., 2019. Earthquake early warning: Advances, scientific challenges, and societal needs. *Annual Review of Earth and Planetary Sciences* 47, 361–388. doi:10.1146/annurev-earth-053018-060457.
- Allen, R.V., 1978. Automatic earthquake recognition and timing from single traces. *Bulletin of the Seismological Society of America* 68, 1521–1532.
- Bahdanau, D., Cho, K., Bengio, Y., 2015. Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473 .
- Bai, L., Yao, L., Li, C., Wang, X., Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting. *Advances in Neural Information Processing Systems* 33, 17804–17815.
- Baltrusaitis, T., Ahuja, C., Morency, L.P., 2019. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 423–443.
- Bergen, K.J., Johnson, P.A., de Hoop, M.V., Beroza, G.C., 2019. Machine learning for data-driven discovery in solid Earth geoscience. *Science* 363, eaau0323.
- Beroza, G.C., Segou, M., Mousavi, S.M., 2021. Machine learning and earthquake forecasting—next steps. *Nature Communications* 12, 4761. doi:10.1038/s41467-021-24952-6.
- Bilal, M.A., Ji, Y., Wang, Y., Akhtar, M., Maqsood, A., 2022. Early earthquake detection using graph neural networks on seismometer network data. *Sensors* 22, 7434.
- Bloemheugel, S., van den Hoogen, J., Jozinovic, D., Micheline, A., 2022. Graph neural networks for multivariate time series regression with application to seismic data. *International Journal of Data Science and Analytics* 14, 317–332.
- Bommasani, R., Hudson, D.A., Adeli, E., Altman, R., Arber, S., von Arx, S., et al., 2021. On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258 .
- Boore, D.M., Stewart, J.P., Seyhan, E., Atkinson, G.M., 2014. NGA-West2 equations for predicting PGA, PGV, and 5% damped PSA. *Earthquake Spectra* 30, 1057–1085.
- Bose, M., Heaton, T.H., Hauksson, E., 2018. FinDer v.2: Improved real-time ground-motion predictions for earthquake early warning with a new reference data set. *Bulletin of the Seismological Society of America* 108, 2904–2919.
- Bronstein, M.M., Bruna, J., LeCun, Y., Szegedy, A., Vandergheynst, P., 2017. Geometric deep learning: going beyond Euclidean data. *IEEE Signal Processing Magazine* 34, 18–42.
- Chen, J., Ran, X., 2019. Deep learning with edge computing: A review. *Proceedings of the IEEE* 107, 1655–1674.
- Chen, T., Kornblith, S., Norouzi, M., Hinton, G., 2020. A simple framework for contrastive learning of visual representations. *International Conference on Machine Learning* , 1597–1607.

- Clements, T., Cochran, E.S., Baltay, A.S., Minson, S.E., Yoon, C.E., 2024. GRAPES: Earthquake early warning by passing seismic vectors through the grapevine. *Geophysical Research Letters* 51, e2023GL107389. doi:10.1029/2023GL107389.
- Cuturi, M., Blondel, M., 2017. Soft-DTW: a differentiable loss function for time-series. *International Conference on Machine Learning*, 894–903.
- Defferrard, M., Bresson, X., Vandergheynst, P., 2016. Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in Neural Information Processing Systems* 29.
- Derras, B., Bard, P.Y., Cotton, F., 2014. Towards fully data driven ground-motion prediction models for Europe. *Bulletin of Earthquake Engineering* 12, 495–516.
- Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding, in: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pp. 4171–4186.
- DeVries, P.M.R., Viegas, F., Wattenberg, M., Meade, B.J., 2018. Deep learning of aftershock patterns following large earthquakes. *Nature* 560, 632–634.
- Dhir, K., Singh, P., 2026. Next-gen earthquake monitoring: leveraging generative AI and deep learning for early warning and response. *Journal of Seismology* 30. doi:10.1007/s10950-026-10370-y.
- Dwivedi, V.P., Bresson, X., 2021. A generalization of transformer networks to graphs. *arXiv preprint arXiv:2012.09699*.
- Ellsworth, W.L., 2013. Injection-induced earthquakes. *Science* 341, 1225942.
- van den Ende, M.P., Ampuero, J.P., 2020. Automated seismic source characterization using deep graph neural networks. *Geophysical Research Letters* 47, e2020GL088690.
- van den Ende, M.P., Ampuero, J.P., 2022. Deep deconvolution for earthquake source characterization with graph neural networks. *Geophysical Journal International* 232, 2120–2131.
- Espinosa-Aranda, J.M., Jimenez, A., Ibarrola, G., Alcantar, F., Aguilar, A., Inostroza, M., Maldonado, S., 1995. The Mexico City seismic alert system. *Seismological Research Letters* 66, 42–53.
- Gasparini, P., Manfredi, G., Zschau, J., 2007. *Earthquake early warning systems*. Springer.
- Gibbons, S.J., Ringdal, F., 2006. The detection of low magnitude seismic events using array-based waveform matched filtering. *Geophysical Journal International* 165, 149–166.
- Gilmer, J., Schoenholz, S.S., Riley, P.F., Vinyals, O., Dahl, G.E., 2017. Neural message passing for quantum chemistry, in: *International Conference on Machine Learning*, pp. 1263–1272.
- Given, D.D., Allen, R.M., Baltay, A.S., Bodin, P., Cochran, E.S., Creager, K., 2018. ShakeAlert: An earthquake early warning system for the United States west coast. *Science Advances* 4.

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y., 2014. Generative adversarial nets, in: Advances in Neural Information Processing Systems (NeurIPS).
- Graves, A., Mohamed, A.r., Hinton, G., 2013. Speech recognition with deep recurrent neural networks. IEEE International Conference on Acoustics, Speech and Signal Processing , 6645–6649.
- Gualandi, A., Avouac, J.P., Galetzka, J., Genrich, J.F., Blewitt, G., Adhikari, L.B., 2017. Pre- and post-seismic deformation related to the 2015, Mw7.8 Gorkha earthquake, Nepal. Tectonophysics 714, 90–106.
- Guo, S., Lin, Y., Feng, N., Song, C., Wan, H., 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. Proceedings of the AAAI Conference on Artificial Intelligence 33, 922–929.
- Hamilton, W., Ying, Z., Leskovec, J., 2017. Inductive representation learning on large graphs, in: Advances in Neural Information Processing Systems, pp. 1024–1034.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 770–778.
- Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Computation 9, 1735–1780.
- Hoshiba, M., Ozaki, T., 2015. Earthquake early warning and tsunami warning of JMA and their performance in the 2011 Tohoku earthquake. Early Warning for Geological Disasters , 1–28.
- Hsiao, N.C., Wu, Y.M., Shin, T.C., Zhao, L., Teng, T.L., 2009. Development of earthquake early warning system in Taiwan. Geophysical Research Letters 36.
- Ioffe, S., Szegedy, C., 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift. International Conference on Machine Learning , 448–456.
- Johnson, P.A., Rouet-Leduc, B., Dascher-Cousineau, K., Brodsky, E.E., Marone, C., 2021. Laboratory earthquake forecasting: A machine learning competition. Proceedings of the National Academy of Sciences 118, e2011362118.
- Jozinović, D., Lomax, A., Stajduhar, I., Michelini, A., 2022. Transfer learning applied to ground motion prediction with graph neural networks. Frontiers in Earth Science 10, 858200.
- Jozinovic, D., Lomax, A., Stajduhar, I., Michelini, A., 2022. Transfer learning based ground motion prediction with graph neural networks. Journal of Geophysical Research: Solid Earth 127, e2022JB024210.
- Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization, in: International Conference on Learning Representations (ICLR).
- Kingma, D.P., Welling, M., 2014. Auto-encoding variational Bayes. arXiv preprint arXiv:1312.6114 .

- Kipf, T.N., Welling, M., 2017. Semi-supervised classification with graph convolutional networks, in: International Conference on Learning Representations (ICLR).
- Kong, Q., Trugman, D.T., Ross, Z.E., Bianco, M.J., Meade, B.J., Gerstoft, P., 2019. Machine learning in seismology: Turning data into insights. *Seismological Research Letters* 90, 681–691.
- Kriegerowski, M., Petersen, G.M., Vasyura-Bathke, H., Ohrnberger, M., 2019. A deep convolutional neural network for localization of clustered earthquakes based on multistation full waveforms. *Seismological Research Letters* 90, 510–516.
- Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. ImageNet classification with deep convolutional neural networks, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. *Nature* 521, 436–444.
- Lee, J., Lee, I., Kang, J., 2019. Self-attention graph pooling. *International Conference on Machine Learning*, 3734–3743.
- Leema, A., Balakrishnan, P., Kiruba, G.G., Rajarajan, G., Goel, S., Aggarwal, P., 2025. Seismo-QuakeGNN: a hybrid framework for spatio-temporal earthquake prediction with transformer-enhanced models. *Frontiers in Artificial Intelligence* 8, 1690476. doi:10.3389/frai.2025.1690476.
- Li, T., Sahu, A.K., Talwalkar, A., Smith, V., 2020. Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine* 37, 50–60.
- Li, Y., Tarlow, D., Brockschmidt, M., Zemel, R., 2016. Gated graph sequence neural networks. *arXiv preprint arXiv:1511.05493*.
- Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: *International Conference on Learning Representations*.
- Liu, T., Münchmeyer, J., Laurenti, L., Marone, C., de Hoop, M.V., Dokmanić, I., 2024. SeisLM: a foundation model for seismic waveforms, in: *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2410.15765.
- Liu, Y., Jin, M., Pan, S., Zhou, C., Zheng, Y., Xia, F., Yu, P.S., 2022. Graph self-supervised learning: A survey. *IEEE Transactions on Knowledge and Data Engineering* 35, 5879–5900.
- Lomax, A., Michelini, A., Curtis, A., 2019. An investigation of rapid earthquake characterization using single-station waveforms and a convolutional neural network. *Seismological Research Letters* 90, 517–529.
- Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* 30.
- Lyu, D., Nakata, R., Ren, P., Mahoney, M.W., Pitarka, A., Nakata, N., Erichson, N.B., 2025. Rapid wavefield forecasting for earthquake early warning via deep sequence to sequence learning. *Nature Communications* 16, 10247. doi:10.1038/s41467-025-65435-2.

- Magrini, F., Jozinovic, D., Cammarano, F., Michelini, A., Boschi, L., 2020. Local earthquakes detection: A benchmark dataset of 3-component seismograms built on a global scale. *Artificial Intelligence in Geosciences* 1, 1–10.
- McBrearty, I.W., Beroza, G.C., 2023a. Earthquake location and magnitude estimation with graph neural networks. *Journal of Geophysical Research: Solid Earth* 128, e2022JB025349.
- McBrearty, I.W., Beroza, G.C., 2023b. Earthquake phase association with graph neural networks. *Bulletin of the Seismological Society of America* 113, 524–547. doi:10.1785/0120220182.
- McBrearty, I.W., Beroza, G.C., 2025. Double difference earthquake location with graph neural networks. *Earth, Planets and Space* 77, 120. doi:10.1186/s40623-025-02251-4.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A., 2017. Communication-efficient learning of deep networks from decentralized data. *Artificial Intelligence and Statistics* , 1273–1282.
- Michelini, A., Cianetti, S., Gaviano, S., Giunchi, C., Jozinovic, D., Lauciani, V., 2021. INSTANCE—the Italian seismic dataset for machine learning. *Earth System Science Data* 13, 5509–5544.
- Mignan, A., Broccardo, M., 2019. One neuron versus deep learning in aftershock prediction. *Nature* 574, E1–E3.
- Morris, C., Ritzert, M., Fey, M., Hamilton, W.L., Lenssen, J.E., Rattan, G., Grohe, M., 2019. Weisfeiler and Leman go neural: Higher-order graph neural networks, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 4602–4609.
- Moseley, B., Markham, A., Nissen-Meyer, T., 2020. Solving the wave equation with physics-informed deep learning. *arXiv preprint arXiv:2006.11894* .
- Mousavi, S.M., Beroza, G.C., 2022. Deep-learning seismology. *Science* 377, eabm4470.
- Mousavi, S.M., Beroza, G.C., 2023. Machine learning in earthquake seismology. *Annual Review of Earth and Planetary Sciences* 51, 105–129.
- Mousavi, S.M., Ellsworth, W.L., Zhu, W., Chuber, L.Y., Beroza, G.C., 2020. Earthquake transformer—an attentive deep-learning model for simultaneous earthquake detection and phase picking. *Nature Communications* 11, 3952.
- Mousavi, S.M., Sheng, Y., Zhu, W., Beroza, G.C., 2019. STanford EArthquake Dataset (STEAD): A global data set of seismic signals for AI. *IEEE Access* 7, 179464–179476.
- Münchmeyer, J., Bindi, D., Leser, U., Tilmann, F., 2022. Transformer earthquake alerting model: A new versatile approach to earthquake early warning. *Geophysical Journal International* 228, 468–486.
- Munchmeyer, J., Woollam, J., Rietbrock, A., Tilmann, F., 2021. Associating detected seismic phases with earthquakes using graph neural networks. *Seismological Research Letters* 93, 503–514.

- Munchmeyer, J., Woollam, J., Rietbrock, A., Tilmann, F., Lange, D., 2022. Which picker fits my data? A quantitative evaluation of deep learning based seismic pickers. *Journal of Geophysical Research: Solid Earth* 127, e2021JB023499.
- Murshed, R.U., Noshin, K., Zakaria, M.A., Uddin, M.F., Amin, A.F.M.S., Ali, M.E., 2024. Real-time seismic intensity prediction using self-supervised contrastive GNN for earthquake early warning. *IEEE Transactions on Geoscience and Remote Sensing* 62, 1–19. doi:10.1109/TGRS.2024.3373643.
- Nakamura, Y., 1988. On the urgent earthquake detection and alarm system (UrEDAS). *Proceedings of the 9th World Conference on Earthquake Engineering* 7, 673–678.
- Pan, S.J., Yang, Q., 2010. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering* 22, 1345–1359.
- Perol, T., Gharbi, M., Denolle, M., 2018. Convolutional neural network for earthquake detection and location. *Science Advances* 4, e1700578.
- Piriyasatit, S., Kuruoglu, E.E., Ozeren, M.S., 2025. Spatio-temporal graph structure learning for earthquake detection. *arXiv preprint arXiv:2503.11215* doi:10.48550/arXiv.2503.11215.
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* 378, 686–707.
- Rasht-Behesht, M., Huber, C., Shukla, K., Karniadakis, G.E., 2022. Physics-informed neural networks for earthquake detection and location. *Journal of Geophysical Research: Solid Earth* 127, e2021JB023158.
- Ross, Z.E., Meier, M.A., Hauksson, E., Heaton, T.H., 2018. Generalized seismic phase detection with deep learning. *Bulletin of the Seismological Society of America* 108, 2894–2901.
- Ross, Z.E., Trugman, D.T., Hauksson, E., Shearer, P.M., 2019a. Searching for hidden earthquakes in Southern California. *Science* 364, 767–771.
- Ross, Z.E., Yue, Y., Meier, M.A., Hauksson, E., Heaton, T.H., 2019b. PhaseLink: A deep learning approach to seismic phase association. *Journal of Geophysical Research: Solid Earth* 124, 856–869.
- Rouet-Leduc, B., Hulbert, C., Lubbers, N., Barros, K., Humphreys, C.J., Johnson, P.A., 2017. Machine learning predicts laboratory earthquakes. *Geophysical Research Letters* 44, 9276–9282.
- Samek, W., Montavon, G., Lapuschkin, S., Anders, C.J., Muller, K.R., 2021. Explaining deep neural networks and beyond: A review of methods and critical assessment. *Proceedings of the IEEE* 109, 247–278.
- Satriano, C., Wu, Y.M., Zollo, A., Kanamori, H., 2011. Earthquake early warning: Concepts, methods, and physical grounds. *Soil Dynamics and Earthquake Engineering* 31, 106–118.
- Scarselli, F., Gori, M., Tsoi, A.C., Hagenbuchner, M., Monfardini, G., 2009. The graph neural network model. *IEEE Transactions on Neural Networks* 20, 61–80.

- Schuster, M., Paliwal, K.K., 1997. Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing* 45, 2673–2681.
- Shi, W., Cao, J., Zhang, Q., Li, Y., Xu, L., 2016. Edge computing: Vision and challenges. *IEEE Internet of Things Journal* 3, 637–646.
- Si, X., Wu, X., Li, Z., Wang, S., Zhu, J., 2024. An all-in-one seismic phase picking, location, and association network for multi-task multi-station earthquake monitoring. *Communications Earth & Environment* 5, 22. doi:10.1038/s43247-023-01188-4.
- Simons, M., Minson, S.E., Sladen, A., Ortega, F., Jiang, J., Owen, S.E., Meng, L., Ampuero, J.P., Wei, S., Chu, R., et al., 2011. The 2011 magnitude 9.0 Tohoku-Oki earthquake: Mosaicking the megathrust from seconds to centuries. *Science* 332, 1421–1425.
- Smith, J.D., Ross, Z.E., Azizzadenesheli, K., Muir, J.B., 2022. HypoSVI: Hypocentre inversion with Stein variational inference and physics informed neural networks. *Geophysical Journal International* 228, 698–710.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research* 15, 1929–1958.
- Tan, Y.J., Waldhauser, F., Ellsworth, W.L., Zhang, M., Zhu, W., Michele, M., Chiaraluce, L., Beroza, G.C., Segou, M., 2021. Machine-learning-based high-resolution earthquake catalog reveals how complex fault structures were activated during the 2016–2017 central Italy sequence. *The Seismic Record* 1, 11–19.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- Velickovic, P., 2022. Message passing all the way up. *arXiv preprint arXiv:2202.13853* .
- Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y., 2018. Graph attention networks, in: *International Conference on Learning Representations*.
- Wang, R., Yin, J., Gupta, S., 2021. Graph neural network for induced seismicity monitoring and forecasting. *Computers and Geosciences* 155, 104857.
- Wang, X., Liu, F., Su, R., Wang, Z., Fang, L., Zhou, L., Bai, L., Ouyang, W., 2025. SeisMoLLM: Advancing seismic monitoring via cross-modal transfer with pre-trained large language model. *arXiv preprint arXiv:2502.19960* doi:10.48550/arXiv.2502.19960.
- Weiss, K., Khoshgoftaar, T.M., Wang, D., 2016. A survey of transfer learning. *Journal of Big Data* 3, 1–40.
- Withers, M., Aster, R., Young, C., Beiriger, J., Harris, M., Moore, S., Trujillo, J., 1998. A comparison of select trigger algorithms for automated global seismic phase and event detection. *Bulletin of the Seismological Society of America* 88, 95–106.

- Woollam, J., Munchmeyer, J., Tilmann, F., Rietbrock, A., Lange, D., Bornstein, T., 2022. SeisBench—a toolbox for machine learning in seismology. *Seismological Research Letters* 93, 1695–1709.
- Wu, A., Khan, I., Kwon, Y.W., 2025. A transformer-based real-time earthquake detection framework in heterogeneous environments. *Scientific Reports* 15, 8420. doi:10.1038/s41598-024-82087-2.
- Wu, Y.M., Kanamori, H., 2005. Experiment on an onsite early warning method for the Taiwan early warning system. *Bulletin of the Seismological Society of America* 95, 347–353.
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S., 2021. A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems* 32, 4–24.
- Wu, Z., Pan, S., Long, G., Jiang, J., Chang, X., Zhang, C., 2020. Connecting the dots: Multivariate time series forecasting with graph neural networks. *Proceedings of the 26th ACM SIGKDD*, 753–763.
- Xu, K., Hu, W., Leskovec, J., Jegelka, S., 2019. How powerful are graph neural networks?, in: *International Conference on Learning Representations*.
- Yano, K., Shiina, T., Kurata, S., Kato, A., Komaki, F., Sakai, S., Hirata, N., 2021. Graph-partitioning based convolutional neural network for earthquake detection using a seismic array. *Journal of Geophysical Research: Solid Earth* 126, e2020JB020269. doi:10.1029/2020JB020269.
- Ying, C., Cai, T., Luo, S., Zheng, S., Ke, G., He, D., Shen, Y., Liu, T.Y., 2021. Do transformers really perform bad for graph representation?, in: *Advances in Neural Information Processing Systems*.
- Ying, Z., Bourgeois, D., You, J., Zitnik, M., Leskovec, J., 2019. GNNExplainer: Generating explanations for graph neural networks. *Advances in Neural Information Processing Systems* 32.
- Ying, Z., You, J., Morris, C., Ren, X., Hamilton, W., Leskovec, J., 2018. Hierarchical graph representation learning with differentiable pooling, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- Yoon, C.E., O'Reilly, O., Bergen, K.J., Beroza, G.C., 2015. Earthquake detection through computationally efficient similarity search. *Science Advances* 1, e1501057.
- Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3634–3640.
- Zhang, J., Zhao, B., Zhu, W., 2022a. Graph-based deep learning for seismic event classification. *Geophysical Journal International* 231, 1149–1163.
- Zhang, X., Bergen, K.J., Beroza, G.C., Biondi, B., Zhu, W., 2021. Spatiotemporal graph convolutional networks for earthquake source characterization. *Journal of Geophysical Research: Solid Earth* 126, e2021JB022610.
- Zhang, X., Zhang, M., 2024. Universal neural networks for real-time earthquake early warning trained with generalized earthquakes. *Communications Earth & Environment* 5, 528. doi:10.1038/s43247-024-01718-8.

- Zhang, Z., Cui, P., Zhu, W., 2022b. Deep learning on graphs: A survey. *IEEE Transactions on Knowledge and Data Engineering* 34, 249–270.
- Zhao, L., Song, Y., Zhang, C., Liu, Y., Wang, P., Lin, T., Deng, M., Li, H., 2020. T-GCN: A temporal graph convolutional network for traffic prediction. *IEEE Transactions on Intelligent Transportation Systems* 21, 3848–3858.
- Zhou, J., Cui, G., Hu, S., Zhang, Z., Yang, C., Liu, Z., Wang, L., Li, C., Sun, M., 2020. Graph neural networks: A review of methods and applications. *AI Open* 1, 57–81.
- Zhu, W., Beroza, G.C., 2019. PhaseNet: A deep-neural-network-based seismic arrival-time picking method. *Geophysical Journal International* 216, 261–273.
- Zhu, W., Hou, A.B., Yang, R., Datta, A., Mousavi, S.M., Ellsworth, W.L., Beroza, G.C., 2023a. Quake-Flow: A scalable machine-learning-based earthquake monitoring workflow with cloud computing. *Geophysical Journal International* 232, 684–693.
- Zhu, W., McBrearty, I.W., Mousavi, S.M., Ellsworth, W.L., Beroza, G.C., 2022. Earthquake phase association using a Bayesian Gaussian mixture model. *Journal of Geophysical Research: Solid Earth* 127, e2021JB023249.
- Zhu, W., Mousavi, S.M., Beroza, G.C., 2023b. Seismic arrival-time picking on distributed acoustic sensing data using semi-supervised learning. *Nature Communications* 14, 994.
- Zollo, A., Amoroso, O., Lancieri, M., Wu, Y.M., Kanamori, H., 2014. Earthquake early warning system in southern Italy. *Geophysical Research Letters* 37.